

# Automatic Groups

And Biautomatic Structures of Geometrizable 3 and 4-Manifolds

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This monograph is hosted at the following URL:

<https://github.com/geometricgrouptheorydev/automatic-groups>

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# Abstract

Automatic groups were introduced in the late eighties, as a generalization of group tables for finite groups to discrete infinite groups, forever linking the theory of regular languages to group theory. Automatic and biautomatic structures of a group are of various interest from an algorithmic and computational viewpoint; they provide methods for carrying computations to a wide class of interesting groups, including hyperbolic groups. Indeed, every automatic group has a solvable word problem and satisfies a quadratic Dehn function while every biautomatic group furthermore has a solvable conjugacy problem.

The first aim of this dissertation is to introduce the theory of regular languages, automatic, and biautomatic groups, including important closure properties. The second aim of this dissertation has to do with 3-manifold fundamental groups, the study of which is of great interest, as 3-manifolds are largely determined by their fundamental groups; in fact, every closed, irreducible, non-spherical 3-manifold is indeed uniquely determined by its fundamental group. Thurston's conjecture, proven by Perelman in 2003, permits us to decompose every closed 3-manifold into pieces admitting one of 8 maximal geometries. Similarly, there are 19 classes of maximal 4-dimensional geometries, and we shall determine which of these geometries admit biautomatic structures, though there are some open problems in the 4-dimensional case.

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# Chapter 1

## Automatic Groups

In this chapter, we introduce the theory automatic and biautomatic groups and prove certain properties of interest to us for the following chapter. This chapter's minimal prerequisites are merely group-theoretical, including group actions, the direct and free products, and the word metric.

The first four sections develop the theory of regular languages and finite state automata required both to define automatic groups and prove the properties of interest to us.

### 1.1 Languages

Much like with ordinary languages, we begin by defining an **alphabet**: merely a finite set with its elements called **letters**. A **word**, or a **string**, of length  $n \in \mathbb{N}$  over an alphabet  $A$  is a map  $\{1, 2, \dots, n\} \rightarrow A$ . When  $n = 0$ , we have the unique **empty word**  $\epsilon$  [1]. The set of words over  $A$  is denoted  $A^*$ , and we write a word  $w \in A^*$  as  $a_1 \dots a_n$  where each  $a_i$  is the image of  $i$  under  $w$ . The set of words of length  $n$  is  $A^n$  and the length of a word  $w$  is denoted  $l(w)$ .

The **concatenation** of two words  $w = a_1 \dots a_n$  and  $v = b_1 \dots b_m$  is  $wv = a_1 \dots a_n b_1 \dots b_m$ . With the operation of concatenation,  $A^*$  forms a monoid with identity  $\epsilon$ , called the **free monoid** with generating set  $A$ . Every map  $f : A \rightarrow M$ , where  $M$  a monoid extends uniquely to a monoid epimorphism  $\pi : A^* \rightarrow M$  where  $\pi(a_1 \dots a_n) = f(a_1) \dots f(a_n)$ . We avoid writing  $\pi$  explicitly and instead write  $w_M$  or  $\bar{w}$  for  $\pi(w)$  and  $v =_M w$  for  $\pi(v) = \pi(w)$ .

A **language** over an alphabet  $A$  is a subset of  $A^*$ . By the paragraph above, given a map  $f : A \rightarrow M$ , then for any language  $L$  over  $A$  we have a well-defined extension  $h : L \rightarrow M$ .

Given an alphabet  $A$ , we also want to consider ordered pairs of words in  $A$  as words over  $A \times A$ . However,  $A^* \times A^* \neq (A \times A)^*$  in general, because the latter set consists only of ordered pairs  $(v, w) \in A^* \times A^*$  with  $v, w$  of the same length. To overcome this difficulty, we add a **padding symbol**  $\$$  to  $A$ ,  $\$ \notin A$  and denote  $A^p = A \cup \{\$\}$ .

**Definition 1.1.1** (Padded pair [2]). *Let  $A$  be an alphabet. Given a pair  $(v, w) \in A^* \times A^*$  we associate a **padded pair**  $(v, w)^p \in (A^p \times A^p)^*$  where the superscript "p" denotes that we append  $\$$  symbols to the shorter of  $v, w$  so that they become of the same length.*

For example, the padded pair of the ordered pair of words  $(group, automaton)$  is  $(group$$$$, automaton) = (g, a)(r, u)(o, t)(u, o)(p, m)(\$, a)(\$, t)(\$, o)(\$, n)$ . The intent of the padded symbol  $\$$  here is nothing more than an indication that the shorter word already finished.

## 1.2 Finite State Automata

For a set  $S$ , we let  $\text{Map}(S, S) = \{f : S \rightarrow S\}$  to be the set of set functions from  $S$  to itself. With the operation of composition,  $\text{Map}(S, S)$  forms a monoid.

**Definition 1.2.1** (Finite State Automaton [3, page 7]). *A (deterministic) **finite state automaton**, or FSA, is a tuple  $(S, A, \mu, Y, s_0)$  where:*

- $S, A$  are the finite sets,  $A$  being the alphabet, and elements of  $S$  called states.
- $Y \subset S$ , whose elements are called **accept states**
- $\mu : A \rightarrow \text{Map}(S, S)$ , called the **transition function**
- $s_0 \in S$  called the **start state** or **initial state**

$\mu$  extends uniquely to a monoid homomorphism  $\mu : A^* \rightarrow \text{Map}(S, S)$ , also denoted  $\mu$ . Furthermore, we may dispose of the usage of  $\mu$  entirely and simply write  $ws$  to mean  $\mu(w)(s)$ . We say that a word  $w \in A^*$  is **recognized/accepted** by the automaton  $M$  if  $ws_0 \in Y$ , and we denote the language of accepted words by  $M$  as  $L(M)$ . We also note that we defined  $\mu$  slightly differently than in [3], where it is defined as a map  $A \times S \rightarrow S$  instead. Both definitions are naturally equivalent.

**Definition 1.2.2.** *A state  $t \in S$  is said to be **accessible** from a state  $s \in S$  if there is a word  $w \in A$  such that  $t = ws$ . States not accessible from  $s_0$  are called **inaccessible states** and a state is **dead** if if no accept state can be accessed from it, otherwise it is a **live state***

**Definition 1.2.3.** *We call a language  $L$  over an alphabet  $A$  a **regular language** if it is the language of accepted words for some FSA  $M$  with alphabet  $A$ .*

**Remark 1.2.4.** *Equivalently, one can define a regular language to be a language defined by a regular expression, though we will not use this definition. See [3, Definition 1.1.3 and Theorem 1.2.7]*

FSA may be depicted as directed graphs [1, page 38], where the vertices are one of the following symbols:

$\odot$                       an accept state

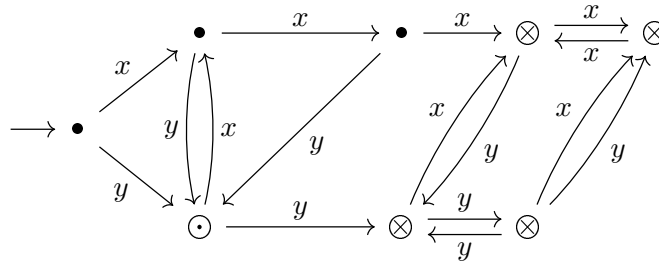
•                      a non – accept live state

$\otimes$                       a dead state

•  $\xrightarrow{a}$  •                      an arrow

$\longrightarrow$  •                      the start state

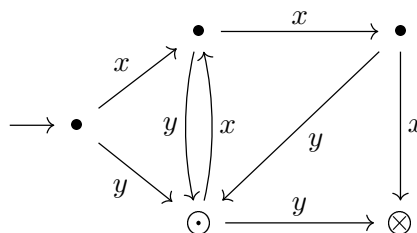
We now consider the following example FSA:



One quickly notices that arrows between dead states give us no information the accepted language. To resolve this redundancy, we will from now on **normalize** our automata.

**Lemma 1.2.6.** *Let  $M'$  be the normalization of an FSA  $M$ . Then  $M'$  is also an FSA and  $L(M') = L(M)$ .*

With normalization, our example automaton becomes:



We also define non-deterministic FSA:

5

the set of start states. We also define  $\mu$  to be a set of arrows from two states in  $S$ , labelled by an element  $A \cup \{\epsilon\}$ ,  $\epsilon \notin A$  representing the empty word.<sup>1</sup>

We define live and dead states as before, and a word  $w \in A^*$  is accepted in  $L$  if there is a sequence of arrows with source  $s_0$  and target an accept state such that the concatenation of all labels is  $w$  (so  $\epsilon$  transitions do not affect  $w$ ). The language of accepted words of a NFA  $N$  is again denoted  $L(N)$ . For  $a \in A \cup \{\epsilon\}$  and  $s \in S$ , we denote by  $as$  the set of states that are the endpoints of an arrow labelled  $a$  starting at  $s$ .

We remark that a deterministic FSA is exactly an NFA where  $S_0$  is a singleton, has no  $\epsilon$  transitions, and for  $s \in S, a \in A$ , there is exactly one arrow labelled  $a$  of source  $s$ . This is because we can then naturally identify the set  $\mu$  with a map  $A \rightarrow \text{Map}(S, S)$ . We sometimes refer to deterministic FSA as DFA to emphasize that they are deterministic.

**Lemma 1.2.8.** *Let  $N$  be an NFA. Then there is a DFA  $M$  such that  $L(N) = L(M)$ . Hence, regular languages are exactly the languages of words accepted by non-deterministic FSA.*

*Proof.* Take  $N = (S, A, \mu, Y, s_0)$  to be an NFA. We follow the strategy of [4, Theorem 2.12] by constructing a deterministic automaton with state set  $P(S)$ , the power set of  $S$ , and start state  $Y$ . A state in  $M$  is deemed an accept state if it contains an element of  $S_0$ . Finally, we define the transition function for  $a \in A, T \in P(S)$  by  $aT = \{s \in S \mid as \cap T \neq \emptyset\}$ . Hence, for any word  $w$  in  $A$ ,  $wY = \{s \in S \mid ws \cap Y \neq \emptyset\}$ . Now we show that  $L(N) = L(M)$ :

Suppose first that  $w \in L(M)$ , so that  $wY$  contains some  $s_0 \in S_0$ . Then  $ws_0 \cap Y \neq \emptyset$ . Hence,  $wY$  is an accept state of  $M$  so that  $L(M) \subset L(N)$ . Conversely, if  $w \in L(N)$ , then  $s_0 \in wY$ , so  $ws_0 \cap Y \neq \emptyset$ . Hence  $w \in L(M)$ , thereby  $L(N) \subset L(M)$ . Therefore,  $L(N) = L(M)$ , so  $L(N)$  is a regular language as required.  $\square$

This lemma is the reason we defined NFA; generally, the smallest NFA accepting a particular language has less states than any DFA accepting it.

## 1.3 Regular Languages

Regular languages enjoy multiple closure properties that will prove to be useful later.

**Lemma 1.3.1.** *For any FSA  $M$  and  $M'$  with alphabet  $A$ , there is an automaton  $N$  such that  $L(N) = L(M) \cap L(M')$ . Hence, the intersection of two regular languages over  $A$  is again a regular language. Similarly, the union of two regular languages is regular.*

*Proof.* Let  $M = (S, A, \mu, Y, s_0)$  and  $M' = (S', A, \mu', Y', s'_0)$ . Then consider the automaton  $N = (T, A, \eta, Z, t_0)$  where  $T = S \times S'$ ,  $t_0 = (s_0, s'_0)$ ,  $Z = Y \times Y'$ , and  $\eta(a)(s, s') = (\mu(a)(s), \mu'(a)(s'))$ . Now clearly for any word  $w \in A^*$ , we have  $ws_0 \in Y$  and  $ws'_0 \in Y'$  iff  $wt_0 = (ws_0, ws'_0) \in Y \times Y'$  hence  $L(N) = L(M) \cap L(M')$ .

By letting  $Z = Y \times S' \cup S \times Y'$ , we obtain an automaton accepting  $L(M) \cup L(M')$ . Alternatively, using NFA, let  $M = (S, A, \mu, Y, s_0)$  and  $M' = (T, B, \eta, Z, t_0)$  be FSA

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<sup>1</sup>More formally, we can define  $\mu$  to be a multivalued function  $A \cup \{\epsilon\} \rightarrow \text{Map}(S, S)$ . Such a viewpoint can be found in [4, Section 2.3.5].

accepting  $L$  and  $K$  respectively, with  $S \cap T = A \cap B = \emptyset$ . Consider the NFA  $P := (S \cup T, A \cup B, \mu \cup \eta, Y \cup Z, s_0, t_0)$  where  $\mu \cup \eta$  is the union of the set of arrows obtained from  $\mu$  and  $\eta$ . Then clearly,  $L(P) = L \cup K$ , because starting at  $s_0$ , transitions can only reach an accept state in  $Y$ , and similarly, starting at  $t_0$ , transitions can only reach an accept state in  $Z$ .  $\square$

**Remark 1.3.2.** *The dead states of the FSA  $N$  accepting the intersection  $L(M) \cap L(M')$  is of from  $(s, s')$  where  $s$  or  $s'$  are dead in their respective automata, so in general  $N$  is not normalized even if  $M$  and  $M'$  are.*

**Lemma 1.3.3.** *Finite languages are regular. In particular,  $A$  is a regular language over itself.*

*Proof.* Now suppose that we have a finite language  $K$ , and consider  $w = w_1 \dots w_n \in K$ . If  $w \neq \epsilon$ , then  $w$  is accepted by the following FSA:

$$\longrightarrow \bullet \xrightarrow{w_1} \bullet \xrightarrow{w_2} \bullet \xrightarrow{\dots} \bullet \xrightarrow{w_n} \odot$$

Then simply take the union of the automata accepting each  $w \in K$ , turning the start state into an accept state if  $\epsilon \in K$ .  $\square$

**Lemma 1.3.4.** *If  $L$  is a regular language, then so is  $A^* \setminus L$*

*Proof.* Let  $M$  be an automaton accepting  $L$ . Turning every non-accept state into and accept state and vice versa yields an automaton accepting  $A^* \setminus L$ .  $\square$

**Definition 1.3.5.** *Let  $K$  and  $L$  be languages over the same alphabet  $A$ . Then their **concatenation** is  $KL = \{vw \in A^* \mid v \in K, w \in L\}$ . We may, of course, concatenate a language with itself, and thus inductively define  $K^n = K^{n-1}K$  for natural  $n$ , where it is understood that  $K^0 = \{\epsilon\}$ . The star closure of  $K$  is then defined  $K^* = \bigcup_{n \geq 0} K^n$ .*

We make a few remarks about these operations. Firstly, the set of languages forms a monoid under concatenation with identity  $\{\epsilon\}$ . Secondly, as  $A$  is a language over  $A$ , we now have two definitions of  $A^*$ ; easily seen to be equivalent, since an element of  $A^n$  is a word of length  $n$ .

**Lemma 1.3.6.** *Let  $K$  and  $L$  be regular languages over an alphabet  $A$ . Then both  $KL$  and  $K^*$  are regular. In particular  $A^*$  is regular over  $A$ .*

*Proof.* We construct NFA accepting  $KL$  and  $K^*$  and appeal to Theorem 1.2.8.

Let  $M = (S_1, A, \mu_1, Y_1, s_1)$  and  $N = (S_2, A, \mu_2, Y_2, s_2)$  be automata accepting  $K$  and  $L$  respectively. We will construct from them an automaton we call  $MN = (S, A, \mu, Y, S_0)$  accepting  $KL$ . Let  $MN$  have start state set  $S_0 = \{s_1\}$  and state set  $S$  the disjoint union of  $S_1$  and  $S_2$ , and the set of accept states  $Y = Y_2$ . We let  $\mu$  have every arrow that  $\mu_1$  and  $\mu_2$  have but we also add arrows labelled  $\epsilon$  from each  $y \in Y_1$  to  $s_2$ . The result is an automaton of shape:

$$M \longrightarrow Y_1 \xrightarrow{\epsilon} s_2 \longrightarrow N$$

Then  $L(MN) = L(M)L(N)$  since any path of arrows from  $s_1$  to  $Y_2$  passes through  $Y_1$  and then  $s_2$  via an  $\epsilon$  transition.



To show that  $K^*$  is regular, we alter the DFA  $M$  accepting  $K$  into an NFA by adding  $\epsilon$  transitions from every member of  $Y$  to  $q$ . We now have a non-deterministic FSA of shape:

$$\longrightarrow s_0 \begin{array}{c} \xrightarrow{\quad} M \longrightarrow Y \\ \qquad\qquad \epsilon \end{array}$$

Then it is clear that any word accepted by this automaton is in some  $K^n, n \geq 0$ , hence  $K^*$  is its set of accept states, as required.  $\square$

We remark that by a simple induction proof, any finite concatenation  $L_1 \dots L_n$  of regular languages is regular, and that an automaton accepting  $L_1 \dots L_n$  has  $\epsilon$  transitions with sources the final states of the automaton accepting  $L_j$  and target the start state of the automaton accepting  $L_{j+1}$ .

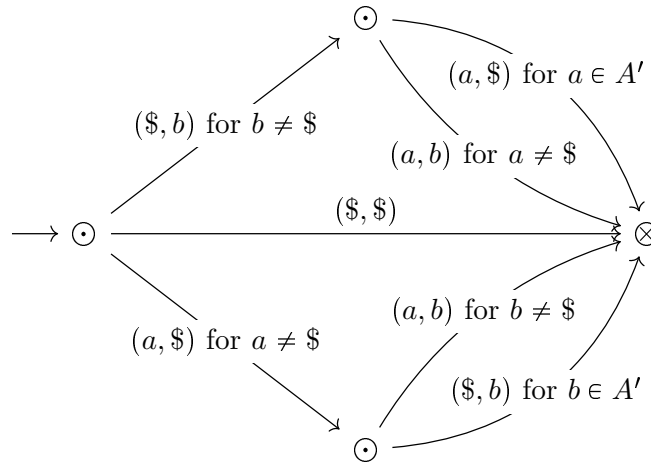
Finally, we notice the following about any regular languages  $K$  and  $L$ :

$$(L \cup K)^{-1} = L^{-1} \cup K^{-1}, (L \cap K)^{-1} = L^{-1} \cap K^{-1}, (LK)^{-1} = K^{-1}L^{-1}, \text{ and } (L^*)^{-1} = (L^{-1})^*$$

## 1.4 Two-Variable Finite State Automata

A (synchronous) two-variable FSA  $M$  over  $A$  is an FSA with alphabet  $A^p \times A^p$  accepting only padded pairs. The following lemma shows that the language of padded pairs is regular:

**Lemma 1.4.1.** *Consider the automaton  $P$  with alphabet  $A^p \times A^p$  given by:*



Here we omit all arrows with the same source and target, so that for any state  $s$ , then  $(a, b)s = s$  whenever  $(a, b)$  satisfies none of the conditions stated by the arrows starting at  $s$ . Then  $P$  accepts  $(v, w) \in (A^p \times A^p)^*$  if and only if it is a padded pair.

*Proof.* Let  $(v, w) \in (A^p \times A^p)^*$  and the initial state be called  $s_0$ . Then we observe that  $ws_0$  is the fail state if and only if at least one of the following occurs:

- \$ appears in both  $v$  and  $w$
- $v$  or  $w$  have a \$ that is followed by any letter  $\neq \$$

Padded pairs, by definition, are precisely the words in  $A^p \times A^p$  that do not satisfy either of these two conditions, so padded pairs are indeed accepted by  $P$ , and any other words in  $A^p \times A^p$  are not.  $\square$

**Corollary 1.4.2.** *For any automaton  $M$  with alphabet  $A^p \times A^p$ , we get a new automaton accepting exactly the padded pairs accepted by  $M$ .*

We can generalize the corollary above:

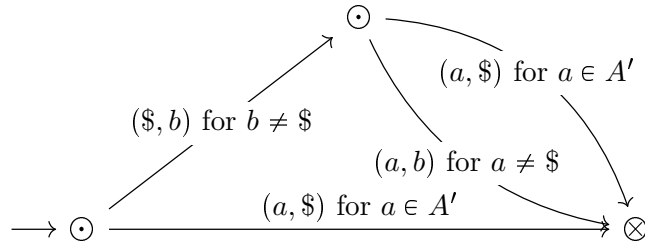
**Lemma 1.4.3.** *Let  $M$  and  $M'$  be FSA with alphabet  $A$ . Then there is a 2-variable FSA  $N$  over  $A$  such that  $L(N) = \{(v, w)^p \mid v \in L(M), w \in L(M')\}$ .*

*Proof.* We let  $S, S'$  be the state sets,  $s_0, s'_0$  be the start states, and  $Y, Y'$  be the set of accept states of  $M, M'$  respectively. We set  $N'$  to be an automaton over  $A^p \times A^p$  with state set  $S \times S'$ , accept state set  $Y \times Y'$ , start state  $(s_0, s'_0)$ , and transitions  $(a, b)(s, s') = (as, bs')$ , where  $\$s = s$  for any  $s \in S$  or  $S'$ . By Corollary 1.4.2, we obtain from  $N'$  an FSA  $N$  that accepts exactly the padded pairs accepted by  $N'$ . By construction,  $L(N) = \{(v, w)^p \mid v \in L(M), w \in L(M')\}$   $\square$

We can also obtain 2-variable FSA with accepted language respecting a partial order.

**Lemma 1.4.4.** *Let  $A$  be an alphabet. Then  $L = \{(v, w)^p : v, w \in A^*, l(v) \leq l(w)\}$  is a regular language over  $A^p \times A^p$*

*Proof.* The following FSA accepts  $L$ , where we omit transitions to and from the same state.



$\square$

From an ordering of an alphabet  $A$ , we give  $A^*$  a **ShortLex** ordering in the following way:

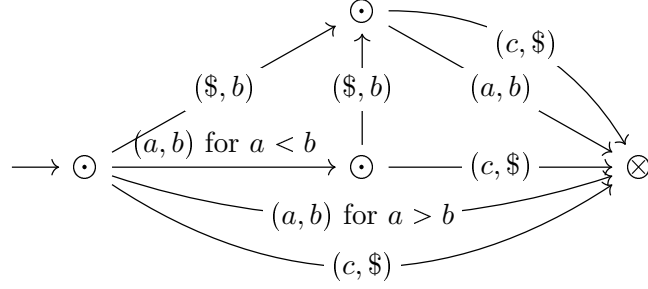
- Obtain a lexicographic ordering in words of the same length in  $A^*$ , where  $a_1 \dots a_n < b_1 \dots b_n$  if  $a_i < b_i$ , where  $i$  is the smallest index such that  $a_i \neq b_i$ .
- A **ShortLex** ordering on  $A^*$  is defined by  $v < w$  if  $l(v) < l(w)$  or  $v$  is lexicographically smaller than  $w$ .

Under the ShortLex ordering, any two words in  $A^*$  are clearly comparable, thus rendering  $A^*$  into a fully ordered set.

**Lemma 1.4.5.** *Let  $\leq$  be any ShortLex ordering on  $A^*$ .*

*Then  $L = \{(v, w)^p : v, w \in A^*, v \leq w\}$  is a regular language over  $A^p \times A^p$ .*

*Proof.*  $L$  is accepted by the following 2-variable FSA, where we let  $a, b$  range over  $A$ ,  $c$  over  $A^p$ :



We again omit transitions to and from the same state.  $\square$

**Lemma 1.4.6.** *Let  $M$  be an FSA over  $A^p \times A^p$ . Then there exists an FSA  $M_{\exists}$  over  $A$  such that  $L(M_{\exists}) = \{w \in A^* : \exists v \in A^*, (v, w)^p \in L(M)\}$ . We therefore also have an FSA  $M_{\nexists}$  with  $L(M_{\nexists}) = \{w \in A^* : \nexists v \in A^*, (v, w)^p \in L(M)\}$ .*

*Proof.* We let  $M_{\exists}$  have the same state set, start state, and accept states as  $M$ . We replace a transition's label  $(a, b)$  with  $b$  if  $b \in A$  or otherwise  $\epsilon$  when  $b = \$$ . By the construction,  $L(M_{\exists}) = \{w \in A^* : \exists v \in A^*, (v, w)^p \in L(M)\}$ . Turning every accept state into non-accept states and vice versa gives  $M_{\nexists}$ .  $\square$

**Corollary 1.4.7.** *For any FSA  $M$  over  $A^p \times A^p$ , the following are regular languages over  $A$ :*

- $\{w \in A^* : \nexists v \in A^*, l(v) \leq l(w) \text{ and } (v, w)^p \in M\}$
- $\{w \in A^* : \nexists v \in A^*, v \leq w \text{ and } (v, w)^p \in M\}$  for any *ShortLex* ordering of  $A^*$ ,

*Proof.* The languages are the complements in  $A^*$  of  $\{w \in A^* : \exists v \in A^*, l(v) \leq l(w) \text{ and } (v, w)^p \in M\}$  and  $\{w \in A^* : \exists v \in A^*, v \leq w \text{ and } (v, w)^p \in M\}$  respectively. Hence, we obtain the required FSA by turning every accept state into a non-accept state and vice versa.  $\square$

## 1.5 Automatic Groups

Let  $G$  be a group and  $X \subset G$ , then we define  $X^{-1} = \{x^{-1} \mid x \in X\}$  and  $X^{\pm 1} = X \cup X^{-1}$ . We say that  $X$  is inverse-closed if  $X = X^{-1}$ . A subset  $Y \subset G$  is a **semigroup generator** of  $G$  if every non-identity element is a product of elements of  $Y$ , while  $Y$  is a **group generator** of  $G$ , denoted  $G = \langle Y \rangle$ , if every element of  $G$  is a product of elements of  $Y^{\pm 1}$ , i.e., if  $Y^{\pm 1}$  is a semigroup generator.

A **rational structure** for  $G$  is a regular language  $L$  over  $A$  such that the map  $L \rightarrow G, w \mapsto w_G$  is surjective. We always set  $\$_G = e$ , the identity.

**Definition 1.5.1** (Automatic Group, [2, page 250]). *Let  $G$  be a group finitely generated by  $X$ ,  $A = X^{\pm 1}$ .  $G$  has an **automatic structure with respect to  $X$**  if there exist FSA  $W$  and  $M_a$  for  $a \in A^p$  such that:*

- $W$  is a rational structure over  $G$  with alphabet  $A$ .

- For each  $a \in A^p$ ,  $M_a$  is a 2-variable FSA over  $A$  and for  $v, w \in A^*$ , its padded pair  $(v, w)^p \in L(M_a)$  if and only if  $v, w \in L(W)$  and  $va =_G w$ .

$W$  is called the **word acceptor**,  $M_a$  the **multiplier** for  $a \in A$ , and  $M_\$$  the **equality recognizer**. A group is called **automatic** if it has an automatic structure. An automatic structure is denoted  $(A, L)$  or  $L$ , where  $A$  is the alphabet and  $L$  is the language accepted by the word acceptor  $W$ . If the  $w$  in the condition for  $W$  is furthermore unique, we call  $(A, L)$  an **automatic structure with uniqueness**.

**Remark 1.5.2.** By definition of  $W_\$,$  we have:  $w \in L(W) \iff (w, w) \in L(M_\$)$ .

Automatic groups are also characterized by the fellow traveller property, whence we recall the word metric. Let  $G$  have monoid generator  $A$ , where  $A = A^{-1}$ . For  $w \in A^*$  denote its length by  $l(w)$ . Then for each  $x \in G$ , we define its **word norm** as  $|x| = \min\{l(w) \mid w \in A^*, w_G = x\}$ , which satisfies the following properties. Let  $x, y \in G$ :

- $|x| = 0 \iff x = e$
- $|x| = |x^{-1}|$
- $|xy| \leq |x| + |y|$

The word norm induces a metric space structure  $d : G \times G \rightarrow \mathbb{R}, d(x, y) = |x^{-1}y|$ , called the **word metric with respect to  $A$** .

We define for  $w = a_1 \dots a_n \in A^*$  and  $i \in \mathbb{N}$ :

$$w(i) = \begin{cases} a_1 \dots a_i & \text{if } 1 \leq i \leq n \\ w & \text{if } i > n \end{cases}$$

We call the  $w(i)$  the **prefixes** of  $w$ . A rational structure  $L$  is called **prefix-closed** if for each  $w \in L$ ,  $L$  also contains  $w$ 's prefixes.

**Definition 1.5.3** (Fellow Traveller Property, [5, page 372]). Let  $G = \langle X \rangle$  with word metric  $d$  with respect to  $A = X^{\pm 1}$ . Then two words  $v, w \in A^*$   **$k$ -fellow travel** if for  $i \geq 1, d(v(i)_G, w(i)_G) \leq k$ .

Now let  $L \subseteq A^*$  be a regular language.  $G$  has the **fellow traveller property with respect to  $L$**  if there exists  $k \in \mathbb{N}$  such that for all  $v, w \in L$  with  $d(v_G, w_G) \leq 1$  then  $v, w$   $k$ -fellow travel.

We remark that when  $G = \langle X \rangle$  has an automatic structure with multipliers  $M_a$ , that  $d(v, w) \leq 1 \iff (v, w)^p \in L(M_a)$  for some  $a \in A^p$ , since:

$$(v, w) \in M_a \iff va =_G w \iff a =_G v^{-1}w \iff d(v_G, w_G) = |v_G^{-1}w_G| \leq 1$$

The last inequality holds because  $a$  is of length 1 or 0 (0 when  $a = \$$ ).

**Theorem 1.5.4.** Let  $G = \langle X \rangle$ ,  $X$  finite,  $A = X \cup X^{-1}$ . Then  $G$  has an automatic structure with respect to  $X$  if and only if there is a rational structure  $L$  of  $G$  with alphabet  $A$  and  $k \in \mathbb{N}$  such that  $G$  has the fellow traveler property with respect to  $L$ .

**Remark 1.5.5.** The last condition may be replaced with: for  $v, w \in L, a \in A$  such that  $va =_G w$ ,  $va$  and  $w$   $k$ -fellow travel. Clearly,  $w$  fellow travels  $v$  if and only if  $w$  fellow travels  $va$ .

*Proof of the "only if" direction.* We follow the same strategy as the proofs of [3, Lemma 2.3.2] and [5, Proposition 3.12], but in more detail.

By Lemma 1.2.6, we may without loss of generality suppose that the automata are normalized. For each  $a \in A^p$ , let  $c_a$  be the number of states of  $M_a$ . Fix  $a$  and suppose now that  $(v, w)^p \in L(M_a)$ , and that  $s_0$  is the initial state of  $M_a$ . Clearly, each  $s_i := (v, w)^p(i)s_0 = (v(i), w(i))^p s_0$  is a live state, so there are minimal length  $\alpha_i, \beta_i \in A^*$  such that  $(\alpha_i, \beta_i)^p s_i \in L(M_a)$ .

For notational simplicity, we let  $\gamma_i = (\alpha_i, \beta_i)^p$ . Now, because of the minimality of  $l(\gamma_i) = \max\{l(\alpha_i), l(\beta_i)\}$ , we cannot have  $\gamma_i(j)s_i = \gamma(j')s_i$  with  $j < j'$ . Otherwise, we may shorten the length of  $\gamma_i$  by discarding  $\gamma_i(j+1), \dots, \gamma_i(j')$  from  $\gamma_i$  and still have  $\gamma_i s_i \in L(M_a)$ , contradicting the minimality of  $l(\gamma_i)$ . Hence, the  $\gamma_i(j)s_i$  are distinct and as there are only  $c_a$  states, we must have that  $l(\gamma) \leq c_a$ , i.e.  $l(\alpha_i), l(\beta_i) \leq c_a$ .

Now,  $(\alpha_i, \beta_i)^p s_i = ((v(i)\alpha_i), (w(i)\beta_i))^p s_0$  so  $((v(i)\alpha_i), (w(i)\beta_i))^p \in L(M_a)$ . Hence,  $v(i)\alpha_i a =_G w(i)\beta_i$  and by rearranging  $\alpha_i a \beta_i^{-1} =_G v(i)^{-1}w(i)$ . By taking the word norm on both sides, we thus have:

$$\begin{aligned} d(v(i)_G, w(i)_G) &= |v(i)_G^{-1}w(i)_G| = |(\alpha_i)_G a (\beta_i)_G^{-1}| \\ &\leq |(\alpha_i)_G| + |a| + |(\beta_i)_G^{-1}| = |(\alpha_i)_G| + |a| + |(\beta_i)_G| \leq 2c_a + 1 \text{ for each } i. \end{aligned}$$

Finally, letting  $c = \max_{a \in A \cup \{\emptyset\}}(c_a)$  and  $k = 2c + 1$ , we find that for  $v, w \in L(W)$ ,  $d(v_G, w_G) \leq 1 \implies \exists a, (v, w)^p \in M_a \implies \forall i, d(v(i)_G, w(i)_G) \leq 2c + 1 = k$ , as required.  $\square$

For the converse, we follow the strategy of [1, Section 5.2] and notice that the fellow traveller property can be expressed as an FSA.

**Definition 1.5.6** (Word Difference Automaton). *Let  $G = \langle X \rangle$  be a group and  $A := X^{\pm 1}$ . Define the **word difference automaton** with accept set  $U$  and constant  $k$ ,  $\text{WD}(\Gamma_k, U)$ , to be a 2-variable FSA over  $A$  with:*

- state set  $\Gamma_k \cup \{\otimes\}$ , where  $\Gamma_k = \{x \in G \mid |x| \leq k\}$  and  $\otimes$  is the failure state
- transition function  $\mu : A^p \times A^p \rightarrow \text{Map}(\Gamma_k, \Gamma_k)$  given by  $\mu(a, b)(x) = y$ , where  $y =_G a^{-1}xb$ , if  $y \in \Gamma_k$  and  $\mu(a, b)(x) = \otimes$  otherwise.
- start state  $e$ , the identity.

If  $U$  is a singleton  $\{u\}$ , we write  $\text{WD}(\Gamma_k, U) = \text{WD}(\Gamma_k, u)$

**Lemma 1.5.7.** *Let  $G$  be as above and consider a word difference automaton of constant  $k$ . Then for all words  $v, w$  and for all  $x \in \Gamma_k$ ,  $(v, w)x$  isn't the fail state if and only if for all  $i \in \mathbb{N}$ ,  $(v(i))^{-1}xw(i) \in \Gamma_k$ , i.e.,  $v, w$   $k$ -fellow travel. In this case,  $(v, w)x =_G v^{-1}xw$ .*

*Proof.* Let  $v = a_1 \dots a_n$  and  $w = b_1 \dots b_n$ . Then by induction we have  $(v, w)x = (a_n, b_n) \dots (a_1, b_1)x = (a_n, b_n) \dots (a_2, b_2)(a_1^{-1}xb_1)_G = (a_n, b_n) \dots (a_3, b_3)(a_2^{-1}a_1^{-1}xb_1b_2)_G = \dots = (a_n^{-1} \dots a_1^{-1}xb_1 \dots b_n)_G$ . At each step, we never reach the fail state as the  $(v(i))^{-1}xw(i) = a_i^{-1} \dots a_1^{-1}xb_1 \dots b_i \in \Gamma_k$  by the lemma's assumption. Hence,  $(v, w)x =_G v^{-1}xw$ , which is not the fail state, as required.  $\square$

Armed with this lemma, we now prove the "if" direction of Theorem 1.5.4:

**Theorem 1.5.8.** *Let  $G = \langle X \rangle$  be a group satisfying the fellow traveler property with respect to a regular language  $L = L(W)$  over  $A := X \cup X^{-1}$ . Then for each  $a \in A^p$ , there exist finite state automata  $M_a$  with alphabet  $A^p \times A^p$  such that  $L(M_a) = \{(v, w)^p \mid v, w \in L, va =_G w\}$ . Hence  $G$  has an automatic structure.*

*Proof.* Let  $k$  be the fellow traveller constant. For each  $a \in A^p$ , we appeal to Lemmas 1.4.3 and 1.3.1 to define an FSA  $M_a$  such that  $L(M_a) = \{(v, w)^p \mid v, w \in L\} \cap L(\text{WD}(\Gamma_k, a))$ . We now check that this is equal to  $\{(v, w)^p \mid v, w \in L, va =_G w\}$ :

Let  $(v, w)^p \in L(M_a)$ . Then in  $\text{WD}(\Gamma_k, a)$ ,  $(v, w)^p e$  is an accept state. Letting  $v^p, w^p \in (A^p)^*$  such that  $(v^p, w^p) = (v, w)^p$ , then  $a =_G (v^p)^{-1} e w^p =_G v^{-1} w$  since  $\$G = e$ . Therefore,  $va =_G w$  so  $L(M_a) \subseteq \{(v, w)^p \mid v, w \in L, va =_G w\}$ .

Now suppose that  $(v, w)^p \in \{(v, w)^p \mid v, w \in L, va =_G w\}$ . By the fellow traveller property,  $\forall i \geq 1, d(v(i)_G, w(i)_G) = |v(i)_G^{-1} e w(i)_G| \leq k$ , so each  $(v(i)^{-1} e w(i))_G \in \Gamma_k$ . Therefore, the previous lemma applies and  $(v, w)^p e =_G (v^p)^{-1} e w^p = v^{-1} w = a$  as  $va =_G w$ . Hence,  $(v, w)^p \in L(\text{WD}(\Gamma_k, a))$ , so  $L(M_a) \supseteq \{(v, w)^p \mid v, w \in L, va =_G w\}$ .

We thus conclude that  $L(M_a) = \{(v, w)^p \mid v, w \in L, va =_G w\}$ . Therefore,  $G$  has the required automatic structure.  $\square$

**Corollary 1.5.9.** *If  $(A, L)$  is an automatic structure on a group  $G$ . Let  $L' \subseteq L$  be regular such that the natural map  $L' \rightarrow G$  is surjective. Then,  $(A, L')$  is also an automatic structure.*

*Proof.* As the fellow traveller property holds for  $L$ , it trivially holds for  $L'$ .  $\square$

**Corollary 1.5.10.** *Let  $G$  be a group with automatic structure  $(A, L)$  and let  $\pi : L \rightarrow G$  be the projection as in the definition of an automatic group. We define  $L'$  to be the words  $w \in L$  that are the smallest in  $\pi^{-1}(x)$  for  $x \in G$ , with respect to the ShortLex ordering  $\leq$ . Then  $(A, L')$  is an automatic structure with uniqueness.*

*Proof.* Recalling the equality recognizer  $M_\$$ , then

$$L' = \{w \in A^* : \nexists v \in A^*, v \leq w \text{ and } (v, w)^p \in M_\$\},$$

which is regular by Corollary 1.4.4. By Corollary 1.5.9, we are done.  $\square$

We state some elementary properties that are of interest. Firstly, if a group has an automatic structure with respect to a generating set, then it is automatic with respect to any other generating set [3, Page 52]. Secondly, the following lemma, from [3, Page 49], will aid in later proofs. The proof we provide here is more detailed.

**Lemma 1.5.11** (Bounded Length Difference). *Let  $G$  be a group with automatic structure  $(A, L)$ . There is  $N \in \mathbb{N}$  such that for any  $w \in L$  and  $x \in G$  with  $d(w_G, x) \leq 1$ , we have:*

- $v \in L$  such that  $v_G = x$  and  $|v| \leq |w| + N$
- If there is some  $v' \in L$ ,  $v'_G = x$  such that  $|v'| > |w| + N$  then there are infinitely many words in  $L$  representing  $x$ .

*Proof.* By Theorem 1.5.4, we may without loss of generality suppose that the multiplier automata  $M_a$  of the automatic structure  $(A, L)$  are of the form constructed in the proof of Theorem 1.5.8. That is, via the constructions in Lemmas 1.4.3 and

1.3.1,  $M_a$  is constructed such that  $L(M_a) = \{(v, w)^p \mid v, w \in L\} \cap L(\text{WD}(\Gamma_k, a))$ ,  $k$  being the fellow traveller constant. We remark, from this construction, that a path in  $M_a$  from the start state to an accept state induces a path in  $\text{WD}(\Gamma_k, a)$  of the same length from its start state to an accept state, and that  $\text{WD}(\Gamma_k, a)$  has less states than  $M_a$ .

We choose  $N$  larger than the number of states in any of  $(A, L)$ 's automata. Let  $w$  and  $x$  as in the lemma's statement, and  $v \in L$  such that  $v_G = x$ . There is  $a \in A \cup \{\$ \}$  such that one of  $(v, w)^p$ ,  $(w, v)^p$  is accepted by the multiplier  $M_a$ . We call the relevant accept state  $y$  for notational convenience.

Now suppose that  $|v| > |w| + N$  and consider the path of transitions from the start state  $s_0$  to  $y$ . After the automaton reads the first  $|w|$  transitions, we are at a live state  $s$ . As there are at least  $N$  transitions remaining, the sub-path starting at  $s$  and ending at  $y$  necessarily visits the same state at least twice. Hence, there is a loop in the path of transitions from  $s$  to  $y$ . From here, one can eliminate the loop and thus shorten  $v$ . If  $|v| > |w| + N$  still, then this process can be repeated until  $|v| \leq |w| + N$ . One could also lengthen  $v$  arbitrarily by repeating the loop arbitrarily many times. Therefore, we have proved both statements.  $\square$

## 1.6 Biautomatic Groups

**Definition 1.6.1.** *Let  $G = \langle X \rangle$ ,  $A = X \cup X^{-1}$ . Then  $G$  has a **biautomatic structure**  $(A, L)$  with respect to  $X$  if both  $(A, L)$  and  $(A, L^{-1})$  are automatic structures.*

Biautomatic groups have differing definitions in the literature, that are unfortunately not quite equivalent. There is, however, a simple condition on  $L$  under which equivalence is guaranteed.

**Theorem 1.6.2.** *Let  $G = \langle X \rangle$ ,  $X$  finite,  $A = X \cup X^{-1}$ ,  $A^p = A \cup \{\$ \}$ . Suppose that  $L$  is a regular language over  $A$  such that the map  $p : L \rightarrow G$ ,  $p(w) = w_G$  is finite-to-one and surjective. Then  $(A, L)$  is a biautomatic structure if and only if  $L$  satisfies the following two-sided fellow traveller property: there is some  $k \in \mathbb{N}$  such that for all  $v, w \in L$  and  $a, b \in A^p$  with  $p(v) = p(awb)$ , then  $v$  and  $awb$   $k$ -fellow travel (or equivalently,  $v$  and  $aw$   $k$ -fellow travel).*

Furthermore, the definition of  $(A, L)$  being a biautomatic structure in the sense of the two-sided fellow traveller property is equivalent to:  $(A, L)$  is automatic and there is some  $\delta \in \mathbb{N}$  such that for any  $v, w \in L$  and  $a \in A^p$  with  $p(v) = p(aw)$ , then  $v$  and  $aw$   $\delta$ -fellow travel.

The assumption that  $p$  is finite-to-one is essential; [3, Lemma 2.5.5] is incorrect, and corrected in [6], where a counterexample can also be found. This necessity boils down to the possibility that the difference in lengths between  $v$  and  $w$  can be arbitrarily large.

**Lemma 1.6.3** ([6]). *Let  $A$  be a finite semigroup generator of a group  $G$ ,  $A = A^{-1}$ , and  $L$  a regular language of  $A$  such that the projection  $p : L \rightarrow G$ ,  $p(w) = w_G$  is finite-to-one. Then for any  $k \in \mathbb{N}$ , there is  $N \in \mathbb{N}$  such that whenever  $d(av(i)_G, w(i)_G) \leq k$  for every  $i \geq 1$ , where  $a \in A \cup \{e\}$ ,  $v, w \in L$ , then  $l(v)$  and  $l(w_G)$  differ by at most  $N$ .*

*Proof.* Let  $l_1 = l(av)$ ,  $l_2 = l(w)$ , and  $n$  be the number of states in an FSA accepting  $L$ . If  $l_1 < l_2$ , then the path given by the suffix  $w' = w_{l_1} \dots w_{l_2}$  of  $w$  lies in the ball of radius  $k$ , centered at  $av$ , of the Cayley graph of  $G$ . We denote this ball by  $B$ . If the path given by  $w$  passes through some  $b \in B$  more than  $n$  times, then we have  $l_1 \leq t < t' \leq l_2$  such that  $w(t) = w(t')$  and  $w(t)s_0 = w(t')s_0$ , where  $s_0$  is the start state.  $w_t \dots w_{t'}$  is then a loop in the FSA, which can be repeated arbitrarily to obtain infinitely many words equal to  $w$  in  $G$ . By hypothesis, this is not possible, so  $w$  passes through each  $b \in B$  at most  $n$  times. Therefore,  $l_2 - l_1 \leq n|B|$ , and a similar argument in the case of  $l_1 > l_2$  yields a similar bound for  $l_1 - l_2$ .  $\square$

We observe that if  $u, w \in L$  are two  $k$ -fellow travelling words with  $u =_G w$  and that  $l(u) = n + l(w)$ ,  $n \geq 0$ , then  $u^{-1}, w^{-1}$   $k + n$ -fellow travel. This is because, for each  $i$ :

$$\begin{aligned} d(u^{-1}(i), w^{-1}(i)) &= d(u(l(u) - i)u^{-1}, w(l(w) - i)w^{-1}), \because u^{-1}(i) =_G u(l(u) - i)u^{-1} \\ &= d(u(l(u) - i), w(l(u) - n - i)), \because u =_G w \\ &\leq d(u(l(u) - i), w(l(u) - i)) + d(w(l(u) - i), w(l(u) - n - i)) \leq k + n \end{aligned}$$

We are now in position to prove Theorem 1.6.2:

*Proof of Theorem 1.6.2.* We begin with the first "if" statement. By taking  $a = \epsilon$ , automaticity of  $(A, L)$  with fellow-traveller constant  $k$  is clear. Now, consider  $v, w \in L$  such that  $p(v^{-1}a) = p(w^{-1})$ , i.e.,  $p(a^{-1}v) = p(w)$ . Then  $a^{-1}v, w$   $k$ -fellow travel so by the lemma the difference between  $l(a^{-1}v)$  and  $l(w)$  differ by at most a constant  $N$ . Reversing the paths and accounting for the length difference, we obtain that  $v^{-1}a$  and  $w^{-1}$   $k + N$ -fellow travel. Therefore,  $(A, L^{-1})$  is also an automatic structure.

Now suppose that  $(A, L)$  is a biautomatic structure and take  $a, b \in A'$ ,  $v, w \in L$  with  $p(v) = p(awb)$ . Let  $u \in L$  such that  $p(u) = p(aw)$ . Then  $p(v) = p(ub)$  so  $v$  and  $u$   $k$ -fellow travel. Moreover,  $p(u^{-1}) = p(wa^{-1})$  so  $u^{-1}$  and  $wa^{-1}$   $k$ -fellow travel. Reversing the paths again, we obtain that  $u$  and  $aw$   $k + N$ -fellow travel. By the triangle inequality,  $v$  and  $aw$   $2k + N + 1$ -fellow travel.

The final statement is obvious in one direction: simply take  $b = \epsilon$ . For the other direction, take  $p(v) = p(awb)$  and  $u \in L$  such that  $p(u) = p(aw)$  again. Then  $v$  and  $u$   $k$ -fellow travel, by automaticity of  $L$ , and  $u$  and  $aw$   $\delta$ -fellow travel so that  $v$  and  $awb$   $k + \delta + 1$ -fellow travel.  $\square$

Just like for automatic groups, any regular subset (projecting surjectively on the group) of a biautomatic structure is biautomatic and there exists one with uniqueness, the proofs being the same as for the automatic case since  $L' \subseteq L$  implies  $(L')^{-1} \subseteq L^{-1}$ . This allows us to use whichever definition we wish to prove that a group is biautomatic, as there are always finite-to-one biautomatic structures if a group is biautomatic in either sense.

**Corollary 1.6.4.** *Every automatic structure  $(A, L)$  of an abelian group  $G$  is a biautomatic structure.*

*Proof.* Let  $v, w \in L, a \in A$  with  $v =_G aw$ . By commutativity,  $v =_G wa =_G aw$  so that both pairs  $v, w$  and  $w, aw$   $k$ -fellow travel for some  $k \in \mathbb{N}$ . By the triangle inequality, then  $v, aw$   $2k$ -fellow travel. By Theorem 1.6.2, we are done.  $\square$



## 1.7 Equivalence of Automatic Structures

Let  $w$  be a word over an alphabet  $A$  and  $t \in [0, \infty)$ . Then we define  $w(t) = w([t])$ , where  $[t]$  is the largest integer  $\leq t$ .

**Definition 1.7.1.** Let  $G = \langle X \rangle$  with word metric  $d$  with respect to  $X$ , and  $A = X \cup X^{-1}$ . Then  $v, w \in A^*$  **asynchronously  $k$ -fellow travel** if there are non-decreasing surjective maps  $t \mapsto t', t \mapsto t'' : [0, \infty) \rightarrow [0, \infty)$  such that:

$$d(v, w) \leq 1 \implies \forall t \geq 1, d(v(t'), w(t'')) \leq k$$

Let  $(A, L)$  be a rational structure such that for some  $k > 0$ , any  $v, w \in L$  with  $d(v, w) \leq 1$  asynchronously  $k$ -fellow travel. Then we say that  $(A, L)$  is an **asynchronous automatic structure** of  $G$ . We can similarly define asynchronous biautomatic structures.

Two rational structures  $L$  and  $K$  are equivalent, denoted  $L \sim K$ , if there is some  $k > 0$  such that any word  $w \in L$  asynchronously  $k$ -fellow travels  $v \in K$  when  $v =_G w$ . We note that, when  $L, K$  are asynchronous automatic structures,  $L \sim K$  if and only if  $L \cup K$  is an asynchronous automatic structure. We denote the set of equivalence classes of automatic, biautomatic, asynchronous automatic, and asynchronous biautomatic structures by  $S\mathfrak{U}(G)$ ,  $BS\mathfrak{U}(G)$ ,  $\mathfrak{U}(G)$ , and  $B\mathfrak{U}(G)$  respectively.

We immediately obtain that any subset of a (bi)automatic structure  $L$  that is also a (bi)automatic structure is equivalent to  $L$  as their union is just  $L$ . In particular,  $BS\mathfrak{U}(G)$  and  $B\mathfrak{U}(G)$  do not depend on the chosen definition of biautomaticity, as every equivalence class then contains a finite-to-one structure.

We now discuss a property of subsets of  $G$  that is invariant under equivalence of rational structures. Let  $L$  be a rational structure on  $G$ , then  $S \subseteq G$  is  **$L$ -rational** if  $L_S := \{w \in L \mid w_G \in S\}$  is a regular language. We state the following lemmas:

**Lemma 1.7.2** ([7, Page 449]). If  $L' \sim L$  are equivalent rational structures of  $G$ , then any  $S \subseteq G$  is  $L$ -rational if and only if it is  $L'$ -rational.

**Lemma 1.7.3** ([?, Pages 2-3]). Let  $L$  be an asynchronous automatic structure on  $G$  and  $S \subseteq G$   $L$ -rational. Then  $\forall g \in G$ ,  $Sg$  is  $L$ -rational, with  $gS$  also  $L$ -rational when  $L$  is a biautomatic structure.

**Corollary 1.7.4.** Let  $H \leq G$  of finite-index. If  $L$  is an automatic structure of  $G$ , then the right cosets  $Hg$  are  $L$ -rational. If  $L$  is a biautomatic structure, then the two-sided cosets  $gHg'$  are  $L$ -rational.

The proofs of these lemmas require the notion of asynchronous two-tape automata, with which we can also give an equivalent definition of asynchronous automaticity [3, Chapter 7]. We have decided not to define those here due to a lack of space.

## 1.8 Closure Properties

### 1.8.1 Direct and Free Products

In this section, we prove that the direct and free products of two (bi)automatic groups is also (bi)automatic, and some consequences.

We recall the following elementary lemma:

**Lemma 1.8.1.** *Let  $G, H$  be generated by  $X, Y$  respectively. Then  $G \times H$  is generated by the disjoint union  $X \cup Y$ . Consider the word metric on this generating set, then for  $g_1, g_2 \in G$  and  $h_1, h_2 \in H$ , we have:*

$$d(g_1 h_1, g_2 h_2) \leq d(g_1, g_2) + d(h_1, h_2)$$

*Proof.* We simply use the fact that for  $g \in G$  and  $h \in H$ ,  $g$  and  $h$  commute in  $G \times H$

$$d(g_1 h_1, g_2 h_2) = |h_1^{-1} g_1^{-1} g_2 h_2| = |g_1^{-1} g_2 h_1^{-1} h_2| \leq |g_1^{-1} g_2| + |h_1^{-1} h_2| = d(g_1, g_2) + d(h_1, h_2)$$

□

**Theorem 1.8.2.** *Let  $G$  and  $H$  be automatic or biautomatic groups. Then  $G \times H$  is automatic or biautomatic, respectively.*

*Proof.* Let  $G$  and  $H$  have automatic structures  $(A, L)$  and  $(B, K)$ , respectively, with fellow traveller constant  $c$  and  $k$  respectively. We will further insist that these are automatic structures with uniqueness, and that  $A, B$  are disjoint. We now claim that  $(A \cup B, LK)$  is an automatic structure of  $G \times H$ , with fellow traveller constant  $c + k$ .

We already know that  $A \cup B$  generates  $G \times H$ , and that every element of  $G \times H$  may be written in the form  $xy$  where  $x \in G, y \in H$ . Therefore, the projection  $LK \rightarrow G \times H$  is surjective, thereby  $LK$  is a rational structure. Now consider the words  $u_1, u_2 \in LK$  with  $d(u_1, u_2) \leq 1$  in the word metric. We let  $u_j = v_j w_j$ ,  $j = 1, 2$  where each  $v_j \in L, w_j \in K$ . To determine if  $u_1$  and  $u_2$  fellow travel, we must consider a few different cases.

Suppose first that  $u_1 b = u_2$  where  $b \in B \cup \{\epsilon\}$ . Then  $v_1 w_1 b =_G v_2 w_2$  from which we deduce that  $v_1 =_G v_2$  and  $w_1 b =_G w_2$ , by definition of the direct product. By the uniqueness of representatives in the automatic structures, we then have  $v_1 = v_2$ . Therefore,

$$d(u_1(i), u_2(i)) \leq d(v_1(i), v_2(i)) + d(w_1(i - l), w_2(i - l)) \leq 0 + k$$

where  $l = l(v_1) = l(v_2)$  and it is understood that  $w(j) = \epsilon$  for  $j \leq 0$ .

Now suppose that  $u_1 a = u_2$  where  $a \in A$ . Then  $v_1 w_1 a =_G v_2 w_2$  from which we deduce that  $w_1 =_G w_2$  and  $v_1 a =_G v_2$ , so  $d(v_1, v_2) \leq 1$ . We can without loss of generality suppose that  $l(v_1) > l(v_2)$ . As  $L$  is an automatic structure with uniqueness, Lemma 1.5.11 tells us that  $l(v_1) - l(v_2) \leq N$ , for some constant  $N$ . This can be seen by taking  $w = v_1$  and  $x = (v_2)_G$  in the statement of the lemma. We now compute bounds for  $d(u_1(i), u_2(i))$  in 3 cases:

- When  $i \leq l(v_2)$ , then  $d(u_1(i), u_2(i)) = d(v_1(i), v_2(i)) \leq c$
- When  $l(v_2) < i \leq l(v_1)$ , then:

$$\begin{aligned} d(u_1(i), u_2(i)) &= d(v_1(i)\epsilon, v_2(i)w_2(i - l(v_2))) \leq d(v_1(i), v_2(i)) + d(\epsilon, w_2(i - l(v_2))) \\ &\leq c + |w_2(i - l(v_2))| \leq c + N \end{aligned}$$

since the prefix  $w_2(i - l(v_2))$  has length  $\leq N$ .

- When  $i > l(v_1)$ , then  $u_1(i) = v_1 w_1(j)$  and  $u_2(i) = v_2 w_1(j')$  where  $j \leq j'$ . As  $j' - j \leq N$ , then  $w_1(j)^{-1} w_1(j')$  is a word of length  $\leq N$ . Therefore:

$$d(u_1(i), u_2(i)) \leq d(v_1, v_2) + d(w_1(j), w_1(j')) = d(v_1, v_2) + |w_1(j)^{-1} w_1(j')| \leq c + N$$

We now let  $K = \max(c, k) + N$ . We see that in each case,  $d(u_1(i), u_2(i)) \leq K$ . Hence,  $(G \times H, LK)$  is an automatic structure with fellow traveller constant  $K$ .

If  $(A, L)$  and  $(B, K)$  are furthermore biautomatic structures, we may still insist uniqueness on their structures. We remark that  $(LK)^{-1} = K^{-1}L^{-1}$ , and thus we may use an analogous argument to the above to show that  $(LK)^{-1}$  satisfies the fellow traveller property. Therefore,  $(A \cup B, LK)$  is indeed a biautomatic structure.  $\square$

This theorem and the following lemmas will allow us to conclude that finitely generated abelian groups are biautomatic.

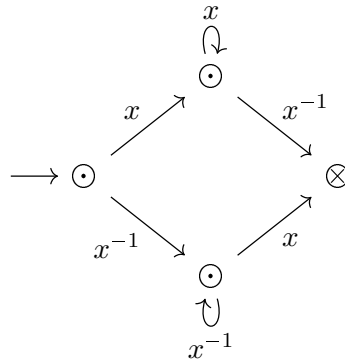
**Lemma 1.8.3.** *Every finite group is biautomatic.*

*Proof.* Let  $G$  be a finite group generated by  $X$  and  $A = X \cup X^{-1}$ . For each  $g \in G$ , choose  $w \in A^*$ ,  $w =_G g$ .  $L$  is then finite, so it is regular, and the fellow traveller property is naturally satisfied as the word metric on a finite group is bounded.  $(A, L^{-1})$  similarly is an automatic structure since the map  $g' \rightarrow G, x \mapsto x^{-1}$  is a bijection.  $\square$

**Lemma 1.8.4.** *The infinite cyclic group is biautomatic.*

*Proof.* Now, let  $C$  be the infinite cyclic group, generated by  $x$ . Then consider the language  $L = \{x\}^* \cup \{x^{-1}\}^*$ , which we know is regular. We remark that the non-empty words in  $L$  are exactly of form  $xx\dots x$  or  $x^{-1}x^{-1}\dots x^{-1}$  with  $x$  or  $x^{-1}$  repeated  $n$  times for any  $n \in \mathbb{N}$ , hence every element of  $C$  has a unique representative in  $L$ . We will denote an element of  $L$  by its unique representative in  $C$  for simplicity. Now, if  $v, w \in L$  and  $d(v, w) = 1$  then either  $vx = w$  or  $wx = v$ . Without loss of generality, let  $vx = w$ . If  $v \in \{x\}^*$  then  $v = x^n$  for some  $n$ , and so  $w = x^{n+1}$ . Then clearly,  $d(v(i), w(i)) \leq 1$  for each  $i$ . If  $w \in \{x^{-1}\}^*$ , we have  $w = x^{-m}$  for some  $m \in \mathbb{N}$ . Then,  $v = x^{-m-1}$ , so once again  $d(v(i), w(i)) \leq 1$  for each  $i$ . Hence,  $(\{x, x^{-1}\}, L)$  is an automatic structure. To see that  $C$  is biautomatic, notice that  $L = L^{-1}$   $\square$

For completeness, we provide an FSA accepting  $L = \{x\}^* \cup \{x^{-1}\}^*$ :



**Corollary 1.8.5.** *Any finitely generated abelian group is biautomatic.*

*Proof.* Every finitely generated abelian group is a finite direct product of cyclic groups, which are biautomatic each.  $\square$

We produce an FSA accepting a biautomatic structure of  $\mathbb{Z}^3$  in Section 2.5.1

**Theorem 1.8.6.** *Let  $G$  and  $H$  be automatic or biautomatic groups. Then  $G * H$  is automatic or biautomatic, respectively.*

*Proof.* We first begin with automaticity. Let  $(A, V)$ ,  $(B, W)$  be automatic structures on  $G$ ,  $H$  respectively, where  $V, W$  are word acceptors with uniqueness. We define a language over  $A \cup B$ ,  $L = (W \cup \{\epsilon\})(VW)^*(V \cup \{\epsilon\})$ , which we know to be regular.

The bijections  $V \rightarrow G$ ,  $W \rightarrow H$  naturally give rise to a bijection  $L \rightarrow G * H$ , since any element of  $G * H$  is of form  $y_1x_1y_2x_2\dots y_nx_n$  for natural  $n$  and for each  $i$ ,  $x_i \in G$ ,  $y_i \in H$ , where we may insist that only  $y_1$  or  $x_n$  may be the identity. Words in  $L$  are in the similar form  $w_1v_1w_2v_2\dots w_nv_n$ ,  $w_i \in W$ ,  $v_i \in V$  where we may again insist that only  $v_1$  or  $w_n$  could be  $\epsilon$ . Ergo, the map  $\pi : L \rightarrow G * H$ ,

$$w_1v_1w_2v_2\dots w_nv_n \rightarrow (w_1)_H(v_1)_G(w_2)_H(v_2)_G\dots(w_n)_H(v_n)_G$$

is indeed a bijection, so  $L$  is a rational structure of  $G * H$ .

We are left with showing the fellow traveller property with respect to  $L$ . We claim that  $L$  satisfies it with constant  $k$ , the maximum of the fellow traveller constants of  $V$  and  $W$ . Let  $d(v, w) \leq 1$  so that  $v = wc$  in  $G * H$  for some  $c \in A \cup B \cup \{\epsilon\}$ . We also know that  $w_G$  is of form  $w = w_1v_1w_2v_2\dots w_nv_n$ ,  $w_i \in W$ ,  $v_i \in V$ , with  $w_1$  or  $v_n$  also possibly the  $\epsilon$ . Then  $v$  written in this form is identical until the last two terms, and follows one of these 3 cases:

- $v = w_1v_1w_2v_2\dots w_nv_nc\epsilon$  if  $c \in B$  and  $v_n \neq \epsilon$
- $v = w_1v_1w_2v_2\dots(w_nc)v_n$  if  $c \in B$  and  $v_n = \epsilon$
- $v = w_1v_1w_2v_2\dots w_n(v_nc)$  if  $c \in A$

In all 3 cases,  $v$  and  $w$  fellow travel since  $w_n$ ,  $w_nc$  and  $v_n$ ,  $v_nc$  fellow travel too. Hence,  $G * H$  is automatic.

For biautomaticity,  $L^{-1} = (V^{-1} \cup \{\epsilon\})(W^{-1}V^{-1})^*(W^{-1} \cup \{\epsilon\})$ . We now repeat the same argument as before, but for  $(A, V^{-1})$ ,  $(B, W^{-1})$  instead.  $\square$

**Remark 1.8.7.** *The theorem above still holds if the product is amalgamated by a finite group. See [3, Theorem 12.1.4]. There are also situations where the above holds for amalgamations for infinite groups. [8] has several such results.*

**Corollary 1.8.8.** *Free groups of finite rank are biautomatic.*

Theorem 1.8.6 has a converse that also applies to semidirect products<sup>2</sup>.

**Theorem 1.8.9.** *Let  $G, H$  be groups. If  $G * H$  or  $G \rtimes H$  is automatic or biautomatic, then  $G$  and  $H$  are automatic or biautomatic respectively.*

*Proof.* As (bi)automaticity is independent of the generating set, we let  $(C, L)$  be a automatic structure for  $G * H$  or  $G \rtimes H$  such that  $C = A \sqcup B$  where  $A = A^{-1}$ ,  $B = B^{-1}$  and  $A, B$  generate  $G, H$  respectively. Let  $L_G = \{w \in L : \bar{w} \in G\}$  and  $L_H = \{w \in L : \bar{w} \in H\}$ . We obtain a DFA accepting  $L_G$  by taking a DFA accepting

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<sup>2</sup>Since if  $G, H$  are generated by  $A, B$  respectively, then  $G \rtimes H$  is generated by  $A \sqcup B$  and  $G, H$  embed in  $G \rtimes H$

$L$  and restricting its transition function to  $A$ , i.e., removing any transitions labelled  $b \in B$ , and similarly for  $L_H$ . Therefore,  $(A, L_G)$  and  $(B, L_H)$  are automatic structures for  $G, H$  respectively.

We notice that  $(L_G)^{-1} = (L^{-1})_G$ . Hence, we argue in the same way that  $(A, (L_G)^{-1})$  and  $(B, (L_H)^{-1})$  are automatic structures when  $(C, L)$  is biautomatic, concluding the biautomatic case.  $\square$

In the case of the free product of biautomatic groups, we can say more:

**Corollary 1.8.10.** *For any two groups  $G, H$ , there is a bijection  $BS\mathfrak{U}(G) \times BS\mathfrak{U}(H) \rightarrow BS\mathfrak{U}(G * H)$*

We defer its proof to [7, Theorem 5.2], as we will not use it.

## 1.8.2 Finite-Index Subgroups

We investigate the relationship between the (bi)automatic structures of a group and a subgroup of finite index.

We recall that, for groups  $H \leq G$ , that left and right transversals are sets of left and right coset representatives, respectively. When  $H$  is of finite index, it is possible to obtain a set that is both a left and right transversal.

**Theorem 1.8.11.** *Let  $G$  be a group with a finite-index subgroup  $H$ . Then  $G$  is (asynchronously) automatic if and only if  $H$  is, with bijections  $S\mathfrak{U}(H) \rightarrow S\mathfrak{U}(G)$  and  $\mathfrak{U}(G) \rightarrow \mathfrak{U}(H)$ .*

*Proof.* The proof is adapted from both [7, Page 447] and [3, Page 89], taking "the best of both worlds".

Suppose first that  $(A, L)$  is an (asynchronous) automatic structure for  $H$ . Let  $P$  be a right transversal of  $H$ , so that any  $x \in G$  can be written in the form  $hp, p \in P, h \in H$ . We claim that  $(A \cup P, LP)$  is an automatic structure<sup>3</sup> of  $G$ .  $LP$  is clearly regular so we only need to show the fellow traveller property:

Let  $w_1, w_2 \in L$ ,  $p_1, p_2 \in P$ , and  $d_G(w_1p_1, w_2p_2) \leq 1$ . Then there is  $a \in A \cup P \cup \{\epsilon\}$  such that  $w_1p_1a =_G w_2p_2$ . Hence,  $(p_1ap_2^{-1})_G = (w_1^{-1}w_2)_G \in L$  and  $d_H(w_1, w_2) = |p_1ap_2^{-1}|_H \leq |p_1|_H + |a|_H + |p_2|_H$ . As there are finitely many possible values for  $p_1, p_2$ , and  $a$ , we deduce that  $d_H(w_1, w_2)$  is bounded. By the fellow traveller property of  $L$ , then the  $d_H(w_1(t'), w_2(t''))$  are uniformly bounded as well, and since for all  $h_1, h_2 \in H$ ,  $d_G(h_1, h_2) \leq d_H(h_1, h_2)$ , the  $d_G(w_1(t'), w_2(t''))$  are uniformly bounded. Therefore, the  $d_G((w_1p_1)(i'), (w_2p_2)(i''))$  are too uniformly bounded.

Now, we show that the above procedure induces a map  $\mathfrak{U}(H) \rightarrow \mathfrak{U}(G)$ . In other words, for automatic structures  $L \sim L'$  and transversals  $P, P'$ , we show that  $LP \sim L'P'$ . To this end, let  $u = vp, u' = v'p'$ , where  $u \in L, u' \in L', p \in P, p' \in P'$  such that  $d(u, u') \leq 1$ . Then once again,  $d(v, v') \leq k$  for some  $k \geq 1$ . As  $L \sim L'$ , then  $L \cup L'$  is an asynchronous automatic structure with fellow traveller constant some  $K \in \mathbb{N}$ . Therefore,  $v$  and  $v'$  asynchronously  $kK$ -fellow travel and so  $u$  and  $u'$   $kK + 2$ -fellow travel. We remark that the same argument as above with  $t = t' = t''$  yields a map  $S\mathfrak{U}(H) \rightarrow S\mathfrak{U}(G)$

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<sup>3</sup>By the argument that follows, we can also conclude that the equivalence class of  $LP$  in  $S\mathfrak{U}(G)$  does not depend on the choice of  $P$ .

Next, we need to construct an inverse map  $\mathfrak{U}(G) \rightarrow \mathfrak{U}(H)$ . Let  $(A, L)$  be an automatic structure of  $G$ , and let  $P$  be simultaneously a left and right transversal of  $H$  in  $G$  with  $e \in P$ . For each  $p \in P$  and  $a \in A$ , we let  $\sigma(p, a)$  be the unique element of  $H$  such that  $pa = \sigma(p, a)q$  for some  $q \in P$ . We notice that  $\sigma(p, a)^{-1} = \sigma(q, a^{-1})$  and define the inverse-closed finite set  $B = \{\sigma(p, a) : p \in P, a \in A\} \subseteq H$ . Let  $w = a_1 \dots a_n \in A^*$ . As  $a_1 = ea_1 = \sigma(e, a_1)p_1$  for some  $p_1 \in P$ , then  $w =_G \sigma(e, a_1)p_1a_2 \dots a_n$ . Then, as  $p_1a_2 = \sigma(p_1, a_2)p_2$  for some  $p_2 \in P$ , then  $w =_G \sigma(e, a_1)\sigma(p_1, a_2)p_2a_3 \dots a_n$ . Repeating this procedure, then

$$w =_G \sigma(e, a_1)\sigma(p_1, a_2)\sigma(p_2, a_3) \dots \sigma(p_{n-1}, a_n)p_n$$

for some  $p_1, \dots, p_n \in P$ . If  $w_G \in H$  then  $p_n = e$ , so  $B$  is a monoid generating set for  $H$ . We let  $L' \subset B^*$  be the words in  $B$  obtained from the procedure above for  $w \in A^*, w_G \in H$ .

To show that the map  $L \mapsto L'$  descends to maps  $\mathfrak{U}(G) \rightarrow \mathfrak{U}(H)$  and  $S\mathfrak{U}(G) \rightarrow S\mathfrak{U}(H)$ , we first need to show that  $(B, L')$  is an automatic structure (the argument for the synchronous case is again almost the same). Let  $M$  be a normalized FSA accepting  $L$  with state set  $S$  and start state  $s_0$ . We define the FSA  $M'$  over  $A$  to have state set  $S \times P$ , start state  $(s_0, e)$ , transitions  $A \rightarrow \text{Map}(S \times P)$ ,  $a(s, p) = (as, q)$ , where  $pa = \sigma(p, a)q$ , and accept states  $(s, e)$  where  $s$  is an accept state of  $M$ . We see that  $L(M') = \{w \in L : w_G \in H\}$ . From  $M'$  we define an FSA  $N$  over  $B$  by replacing every transition labelled  $a$  from state  $(s, p)$  to a transition labelled  $\sigma(p, a)$ , so that  $\sigma(p, a)(s, p) = (as, q)$ . Then  $L(N) = L'$ .

Denoting  $d_A$  and  $d_B$  for the group metrics with respect to generating sets  $A$  and  $B$  respectively, we note that for  $h, h' \in H$ ,  $d_B(h, h') \leq d_A(h, h')$ . Suppose that  $u_1, u_2 \in L'$  with  $d_B(u_1, u_2) \leq 1$  and corresponding words  $v_1, v_2 \in L$  respectively. As  $u_i =_H v_i, i = 1, 2$ , then  $d_A(v_1, v_2)$  is bounded by the maximal word length of an element of  $B$  in terms of  $A$ . Therefore,  $v_1, v_2$  fellow travel in  $d_A$ . Our construction of  $L'$  guarantees that, for each for  $i = 1, 2, t \geq 0$ ,  $u_i(t) = v_i(t)p_{i,t}$  for some  $p_{i,t} \in P$ . Then, for each  $t \geq 0$ ,

$$d_A(u_1(t'), u_2(t'')) = |p_{1,t'}^{-1}v_1(t')^{-1}v_2(t'')p_{2,t''}|_A \leq |p_{1,t'}|_A + d_A(v_1(t'), v_2(t'')) + |p_{2,t''}|_A$$

$P$  is finite, so each term of the right hand side is uniformly bounded, hence  $u_1, u_2$  fellow travel in  $d_A$  and thus in  $d_B$  too. Therefore,  $(B, L')$  is an automatic structure of  $H$ .

For any two automatic structures  $L, K \subseteq A^*$ , we have, by construction,  $(L \cup K)' = L' \cup K'$ , so  $L \sim K$  implies  $L' \sim K'$  by the previous paragraph. Therefore, the map  $L \rightarrow L'$  indeed descends to maps  $\mathfrak{U}(G) \rightarrow \mathfrak{U}(H)$  and  $S\mathfrak{U}(G) \rightarrow S\mathfrak{U}(H)$ . To show that these maps are the sought-after inverses, we notice that  $L'P \sim L$ , immediately seen from the construction of  $L'$ .  $\square$

**Corollary 1.8.12.** *For any virtually abelian group  $G$ ,  $S\mathfrak{U}(G) = \mathfrak{U}(G)$ .*

*Proof.* By the previous theorem, and since every virtually abelian group has a free abelian subgroup of finite index, it suffices to prove the statement for free abelian groups, the proof of which can be found in [7].  $\square$

The situation for biautomatic groups is less known:

**Theorem 1.8.13.** *Let  $H$  be a finite-index subgroup of  $G$ . Then  $H$  is biautomatic if  $G$  is.*

The proof is similar to the automatic case, using Theorem 1.6.2.

In general, even if  $(A, L)$  is a biautomatic structure,  $(A \cup P, LP)$  may not be biautomatic, and it is an open problem whether virtually biautomatic groups are biautomatic in general. It is known, however, that  $G$  is indeed biautomatic when  $H$  is abelian. We construct such a biautomatic structure in Section 1.10.2.

## 1.9 Strongly Geodesically Automatic Groups

This section is mainly due to [1, Sections 5.7, 5.8, and 6.3], with our main goal to prove that hyperbolic groups are biautomatic.

### 1.9.1 Regularity of the Set Of Geodesic Words

Let  $G = \langle X \rangle$ ,  $A = X^{\pm 1}$ . Define  $\mathcal{G}(G, X)$  to be the set of all geodesic words in the word metric over  $X$ , which are the words not equal in  $G$  to any shorter word.

We begin with the following, inspired by [9, Section 5.2].

**Theorem 1.9.1.** *Let  $G = \langle X | r \rangle$  be an infinite group with presentation consisting of a single cyclically reduced relation, then  $\mathcal{G}(G, X)$  is regular.*

*Proof.* Let  $l$  be the smallest integer  $\geq l(r)/2$ . A reduced word  $w$  is geodesic if and only if it doesn't contain a prefix of a conjugate  $s$  of  $r$  of length  $\geq l(s)/2$ .  $s$  is of form  $v^{-1}tv$ ,  $t$  a cyclic permutation of  $r$ . Now,  $l(v) = l(v^{-1})$  so,  $l(s)/2 = l(v) + l(t)/2$ . Therefore,  $w$  is geodesic if and only if it doesn't contain a prefix of a cyclic permutation  $t$  of  $r$  of length  $\geq l$ . ( $l(t) = l(r)$  as  $r$  is cyclically reduced)

We deduce that all reduced words of length  $< l$  are geodesic, and that the following automaton over  $A := X^{\pm 1}$  accepts  $\mathcal{G}(G, X)$ :

- Let the state set  $S$  be the set of reduced words of length  $< l$  with start state the empty word together with the failure state and every state except the failure state an accept state.
- For every  $w \in S$ ,  $a \in A$ , if  $wa \in S$ , there is a transition  $w \xrightarrow{a} aw$ .
- If  $w' \in W$  of length  $l$  can be obtained from  $w$  by deleting its first letter and adding  $a \in A$  at the end, there is a transition  $w \xrightarrow{a} w'$  unless  $wa$  is the first  $l$  letters of a cyclic permutation of  $r$ , in which case the transition targets the failure state instead.
- all remaining arrows target the failure state.

We remark that once the length of the word is  $\geq l$ , the automaton checks the previous  $l$  letters and identifies if they form the first  $l$  letters of a cyclic permutation of  $r$ . Hence, the FSA indeed accepts  $\mathcal{G}(G, X)$ .  $\square$

We provided an explicit construction when  $G$  has a single cyclically reduced generator, and we now use the following definition for the general case.

**Definition 1.9.2.**  *$G$  has the **falsification by fellow traveler property**, or **FFTP**, with respect to  $X$  if  $\forall v \in A^* \setminus \mathcal{G}(G, X) \exists w \in A^*$  such that  $v =_G w$ ,  $l(w) < l(v)$ , and  $v, w$   $k$ -fellow travel.*

**Lemma 1.9.3.** *If  $G$  has the FFTP with respect to  $X$ , then  $\mathcal{G}(G, X)$  is a regular language.*

*Proof.* Let  $G$  have the FFTP with fellow travelling constant  $k$ . We recall that  $(v, w) \in A^p \times A^p$  is accepted by the word difference automaton  $\text{WD}(\Gamma_k, \Gamma_k)$  if and only if  $v, w$   $k$ -fellow travel. Therefore, we use Corollary 1.4.4 to deduce that

$$\mathcal{G}(G, X) = \{w \in A^* : \nexists v \in A^*, l(v) \leq l(w) \text{ and } (v, w)^p \in \text{WD}(\Gamma_k, \Gamma_k)\}$$

is indeed a regular language.  $\square$

**Definition 1.9.4.** *We say that  $G$  is **strongly geodesically automatic** with respect to  $X$  if  $(A, \mathcal{G}(G, X))$  is an automatic structure of  $G$ .*

Thanks to the lemma above, we may determine whether a group is strongly geodesically automatic without verifying directly that  $\mathcal{G}(G, X)$  is regular.

**Theorem 1.9.5.**  *$G$  is strongly geodesically automatic with respect to  $X$  if and only if there is some  $k > 0$  such that any two words  $v, w \in \mathcal{G}(G, X)$   $k$ -fellow travel whenever  $d(v, w) \leq 1$ .*

*Proof.* By Theorem 1.5.4, we need only prove the 'if' case, which amounts to proving that  $\mathcal{G}(G, X)$  is regular. By Lemma 1.9.3 it suffices to show that  $G$  has the FFTP with respect to  $X$ .

Without loss of generality, suppose  $k \geq 2$ . Let  $w \in A^* \setminus \mathcal{G}(G, X)$ . Write  $w = uv$  where  $u$  is the shortest prefix of  $w$  not geodesic, so  $u = tx, t \in \mathcal{G}(G, X), x \in A$ . Letting  $s \in \mathcal{G}(G, X), s =_G u$ . By assumption,  $s$  and  $t$   $k$ -fellow travel. Also,  $|l(s) - l(u)| \leq 2$ . Therefore,  $w = uv$  and  $sv$   $k$ -fellow travel, as  $k \geq 2$ . Since  $l(s) < l(u)$ , then  $l(w) < l(sv)$ . Clearly,  $w =_G sv$  so we're done.  $\square$

**Corollary 1.9.6.** *If  $(A, \mathcal{G}(G, X))$  is an automatic structure for  $G$ , it is furthermore a biautomatic structure.*

*Proof.* This is an immediate consequence of the previous theorem and that  $(\mathcal{G}(G, X))^{-1} = \mathcal{G}(G, X)$ .  $\square$

We remark that a strongly geodesically automatic group can be given a ShortLex geodesic automatic structure as we can take a subset  $\mathcal{G}(G, X)$  minimal with respect to a ShortLex ordering.

## 1.9.2 Hyperbolic Groups

**Definition 1.9.7.** *A group  $G$  finitely generated by  $X$  is hyperbolic if its Cayley graph  $\Gamma(G, X)$  is a hyperbolic metric space, where the metric on  $\Gamma(G, X)$  is the word metric and with all edges having length 1.*

A definition of a hyperbolic metric space can be found in [1, Section 6.1]. Hyperbolicity does not depend on the generating set.

A geodesic bigon in a geodesic metric space  $(\Gamma, d)$  is a pair of points  $x, y \in \Gamma$  with two geodesic paths  $[xy], [xy]'$  joining them.  $\Gamma$  is said to have  **$\delta$ -thin geodesic bigons** if for any geodesic bigon, the paths  $[xy], [xy]'$   $\delta$ -fellow travel. As  $\delta$ -thin geodesic bigons are  $\delta$ -thin geodesic triangles with two equal corner points, hyperbolic metric spaces have  $\delta$ -thin geodesic bigons for some  $\delta > 0$



**Theorem 1.9.8.** *If  $G$  is hyperbolic, then it is strongly geodesically automatic with respect to any generating set  $X$ , with language  $\mathcal{G}(G, X)$ . Hence,  $G$  is also ShortLex geodesic biautomatic with respect to any generating set  $X$ .*

*Proof.* The Cayley graph  $\Gamma(G, X)$  is a hyperbolic metric space, so there is some  $\delta > 0$  such that  $\Gamma(G, X)$  has  $\delta$ -thin bigons. We shall use Theorem 1.9.5.

Let  $v, w \in \mathcal{G}(G, X)$ ,  $d(v, w) \leq 1$ . We need to show that they  $\delta$ -fellow travel, by considering the different cases that may arise. If  $d(v, w) = 0$  then  $v =_g w$ , so  $v, w$  are the paths of a bigon with endpoints the identity and  $v_G$ . Hence,  $v, w$   $\delta$ -fellow travel in this case.

Otherwise,  $d(v, w) = 1$  so  $va =_g w$  for some  $a \in A$ , and  $l(v) - l(w) \in \{-1, 0, 1\}$ .

- If  $l(v) - l(w) = 1$  then  $v$  and  $wa^{-1}$  forms a bigon.
- If  $l(v) - l(w) = -1$  then  $va$  and  $w$  forms a bigon.
- If  $l(v) = l(w)$  then the paths  $v$  and  $w$  each followed by half of the edge labelled  $a$  from with vertices  $v, w$  forms a bigon.

In each of these cases, we see that  $v, w$  then  $\delta$ -fellow travel.  $\square$

We finish this section by stating the following theorem found in [7, Page 452], used in the proof of Theorem 2.6.1.

**Theorem 1.9.9.** *If  $G$  is hyperbolic, then  $S\mathcal{U}(G)$  is a singleton.*

## 1.10 Virtually Central Extensions

### 1.10.1 Group Extensions

We recall some facts about the second cohomology group and its relation with group extensions, as found in [10] and [11, Chapter 6], but using right  $G$ -actions instead of left ones.

**Definition 1.10.1.** *Let  $G$  and  $A$  be groups. Then an **extension of  $G$  by  $A$**  is a short exact sequence:*

$$0 \xrightarrow{i} A \rightarrow E \xrightarrow{\pi} G \rightarrow 0$$

A **based section** of  $\pi$  is a set map  $s : G \rightarrow E$  such that  $s(e) = e$  and  $\pi \circ s$  is the identity.

Two extensions  $0 \rightarrow A \rightarrow E_j \rightarrow G \rightarrow 0$ ,  $j = 1, 2$  are **equivalent** if there is an isomorphism  $E_1 \rightarrow E_2$  making the following diagram commute:

$$\begin{array}{ccccccc}
 & & & E_1 & & & \\
 & & \nearrow & \downarrow & \searrow & & \\
 0 & \longrightarrow & A & & G & \longrightarrow & 0 \\
 & & \searrow & \downarrow & \nearrow & & \\
 & & & E_2 & & & 
 \end{array}$$

Equivalence of group extensions of  $G$  by  $A$  is easily seen to be an equivalence relation.

We will from now on take  $A$  to be an abelian group, with additive notation. Every element of  $E$  can be written in the form  $s(g)i(a)$ ,  $a \in A, g \in G$ . To see this, as  $i$  is injective, we identify  $A$  with  $i(A)$ . Then,  $A = \ker(\pi)$ , so  $A \trianglelefteq E$ .  $\pi$  then descends to an isomorphism  $E/A \cong G$  so that each  $s(g) \in \pi^{-1}(g)$  is a coset representative of  $A$  in  $E$ .

**Lemma 1.10.2.** *The extension  $0 \rightarrow A \xrightarrow{i} E \xrightarrow{\pi} G \rightarrow 0$  induces a **right** group action  $G \rightarrow \text{Aut}(A)$ , denoted  $a \cdot g$  for  $a \in A, g \in G$ , given by  $i(a)s(g) = s(g)i(a \cdot g)$ .*

*Proof.* Identifying  $A$  and  $i(A)$  again, this amounts to verifying that  $a \cdot g = s(g)^{-1}as(g)$  is a right  $G$ -action. Firstly,  $a \cdot e = s(e)^{-1}as(e) = eae = a$ . Now,  $(a \cdot g) \cdot h = (s(g)s(h))^{-1}as(g)s(h)$  and  $a \cdot gh = s(gh)^{-1}as(gh)$ . As both  $s(gh), s(g)s(h) \in \pi^{-1}(gh)$ , we let  $b \in A$  such that  $s(g)s(h) = bs(gh)$ . Then,

$$(a \cdot g) \cdot h = (bs(gh))^{-1}a(bs(gh)) = s(gh)^{-1}as(gh) = a \cdot gh$$

since  $A$  is abelian.  $\square$

Now that we have a group action, we define:

**Definition 1.10.3.** *A **virtually central extension of  $G$  by  $A$**  is an extension  $0 \rightarrow A \rightarrow E \rightarrow G \rightarrow 0$  where the induced action  $G \rightarrow \text{Aut}(A)$  has finite image.*

For each extension  $0 \rightarrow A \xrightarrow{i} E \xrightarrow{\pi} G \rightarrow 0$  and section  $s : G \rightarrow E$ , we define a (set-theoretical) function  $\sigma : G^2 \rightarrow A$  by  $s(g)s(h) = s(gh)i(\sigma(g, h))$ . This is called the **cocycle** determined by  $E$  and  $s$ . The set of cocycles is denoted  $Z^2(G, A)$ , forming an abelian group under pointwise addition. We also notice that for every  $g \in G, \sigma(e, g) = \sigma(g, e) = e$ .

**Lemma 1.10.4.** *The group law in  $E$  is given by*

$$s(g_1)i(a_1)s(g_2)i(a_2) = s(g_1g_2)i(a_1 \cdot g_2 + a_2 + \sigma(g_2, g_1))$$

We simply obtain this formula by direct computation:

$$\begin{aligned} s(g_1)i(a_1)s(g_2)i(a_2) &= s(g_1)s(g_2)i(a_1 \cdot g_2)i(a_2) = s(g_1g_2)i(\sigma(g_2, g_1))i(a_1 \cdot g_2 + a_2) \\ &= s(g_1g_2)i(a_1 \cdot g_2 + a_2 + \sigma(g_2, g_1)) \end{aligned}$$

**Lemma 1.10.5.** *Every  $\sigma \in Z^2(G, A)$  satisfies the cocycle relation: for every  $f, g, h \in G$ ,*

$$\sigma(f, g) \cdot h + \sigma(fg, h) = \sigma(f, gh) + \sigma(g, h)$$

*Conversely, any  $\sigma : G^2 \rightarrow A$  satisfying the cocycle relation and  $\forall g \in G, \sigma(e, g) = \sigma(g, e) = e$  is in  $Z^2(G, A)$ .*

*Proof.* Let  $f, g, h \in G$ , we compute  $s(f)s(g)s(h)$  in the form  $s(fgh)i(a)$  in two different ways and use the associativity of  $E$  to conclude the equality:

$$\begin{aligned} (s(f)s(g))s(h) &= s(fg)i(\sigma(f, g))s(h) = i(\sigma(f, g) \cdot fg)^{-1}s(fg)s(h) \\ &= i(\sigma(f, g) \cdot g^{-1}f^{-1})s(fgh)i(\sigma(g, h)) = s(fgh)i(\sigma(f, g) \cdot g^{-1}f^{-1}fgh)i(\sigma(g, h)) \\ &= s(fgh)i(\sigma(f, g) \cdot h + \sigma(g, h)) \\ s(f)(s(g)s(h)) &= s(f)s(gh)i(\sigma(g, h)) = s(fgh)i(\sigma(f, gh) + \sigma(g, h)) \end{aligned}$$

The converse is also a straightforward computation, where we show that with the group law as in Lemma 1.10.4,  $\sigma$  determines an extension of  $G$  by  $A$ . The cocycle relation ensures associativity of  $E$ .  $\square$

We remark that, by writing every element of  $E$  in the form  $i(a)s(g)$  instead, we obtain a **left** action  $s(g)i(a) = i(g \cdot a)s(g)$ . This approach is much more common, and the actions are related by  $g \cdot a = a \cdot g^{-1}$ . The group law in this case is

$$i(a_1)s(g_1)i(a_2)s(g_2) = i(a_1 + g_1 \cdot a_2 + \psi(g_1, g_2))s(g_1g_2)$$

This formula will not be suitable for the proof of Lemma 1.10.11, due to  $a_2$  being acted on instead of  $a_1$ . Using the right action also slightly changes the cocycle relation from the one of [10, Chapter 6], but the results nonetheless still carry over, so that there is a quotient  $H^2(G, A)$  of  $Z^2(G, A)$  such that  $\sigma, \sigma'$  are in the same equivalence class if and only if they determine equivalent extensions of  $G$  by  $A$ .

## 1.10.2 Virtually Abelian Groups

We recall that a virtually abelian group  $G$  has a finite-index normal free abelian subgroup  $\Lambda$ , so that we obtain a virtually central extension:

$$e \rightarrow \Lambda \rightarrow G \rightarrow F \rightarrow e$$

where  $F$  is finite. To describe a biautomatic structure on  $G$ , we need an automatic structure on  $\Lambda$  that is invariant under the induced action of  $F$  on  $\Lambda$ .

**Lemma 1.10.6.** *Let  $\Lambda, F$  be finitely generated groups, with  $\Lambda$  free abelian, and  $F$  finite. Suppose we have an action  $F \rightarrow \text{Aut}(\Lambda)$ . Then there is a set  $A$  of distinct monoid generators of  $\Lambda$  such that  $A^{-1} = A$ ,  $\forall h \in F, hA = A$ , and there exists a prefix-closed biautomatic structure  $(A, L)$  with uniqueness of  $\Lambda$  consisting of geodesics, with  $\forall h \in F, hL = L$ .*

We defer the (long) proof to [3, Section 4.2]. By the distinctness of the generators, the action by  $H$  restricts to an action on  $A$  by bijections. We extend this action to  $A^*$  so that  $h \cdot a_1 \dots a_n = (h \cdot a_1) \dots (h \cdot a_n)$ ,  $h \in F, a_1 \in A$ . This is the action on  $L$  we refer to in the lemma. We notice that  $(h \cdot w)_\Lambda = h \cdot w_\Lambda$  so the action on  $A^*$  is compatible with the group structure.

**Theorem 1.10.7.** *Let  $G$  be a finitely generated virtually abelian group. Then  $G$  has a prefix-closed biautomatic structure with uniqueness.*

*Proof.* This is clear when  $G$  is finite, so we suppose  $G$  is infinite. We let  $\Lambda \trianglelefteq G$  be finitely generated free abelian. Let  $s$  be a based section of the extension and  $P' = \{s(x) | x \in F, x \neq e\}$ . We use Lemma 1.10.6 to find an automatic structure  $(A, L)$  on  $\Lambda$  invariant under the action induced by  $s$  on  $\Lambda$ .

As in the proof of Theorem 1.8.11, we deduce that  $(A \cup P, L \cup LP')$  is an automatic structure of  $G$ , with prefix-closure and uniqueness obvious. Biautomaticity then easily follows from the invariance of  $L$  under the action by  $F$ . Finally, we remark that [3, Corollary 4.24] furthermore proves that  $L$  consists of geodesic words.  $\square$

## 1.10.3 Regular 2-Cocycles

This section is due to [12]. We let  $0 \xrightarrow{i} A \rightarrow E \xrightarrow{\pi} G \rightarrow 0$  be a virtually central extension given by a regular cocycle  $\sigma \in Z^2(G, A)$ . We state the main result of [12]:

**Theorem 1.10.8.** *Taking  $G$  and  $E$  as in the extension above, we have that  $G$  is biautomatic if and only if  $E$  is.*

We prove one direction of this theorem by constructing a biautomatic structure on  $E$  from a biautomatic structure on  $G$ .

**Definition 1.10.9.** *Let  $G$  be finitely generated by  $X$  and  $L$  an asynchronous automatic structure on  $G$ . Then:*

- $\sigma \in Z^2(X, A)$  is **weakly bounded** if  $\sigma(X, G)$  and  $\sigma(G, X)$  are finite sets.
- $\sigma$  is furthermore  **$L$ -regular** if for all  $a \in A, x \in X$ ,  $\{g \in G : \sigma(g, x) = a\}$  is  $L$ -rational.

An element of  $H^2(X, A)$  is  **$L$ -regular** if it is the image of an  $L$ -regular cocycle by the quotient map  $Z^2(X, A) \rightarrow H^2(X, A)$ .

**Lemma 1.10.10.**

- For all  $L$ -regular cocycles  $\sigma$ ,  $h \in G$ , and  $a \in A$ , the set  $\{g \in G : \sigma(g, h) = a\}$  is  $L$ -rational
- If  $L_1 \sim L_2$  are asynchronous automatic structures, then a cocycle is  $L_1$ -regular if and only if it is  $L_2$ -regular.

*Proof.* As the first statement is, by definition, true for  $h \in X$ , it suffices to show that if the statement is true for  $h_1, h_2 \in G$ , then it is true for  $h = h_1 h_2$ . For any  $\sigma \in Z^2(G, A)$ ,

$$\sigma(g, h_1 h_2) = \sigma(g, h_1) \cdot h_2 + \sigma(gh_1, h_2) - \sigma(h_1, h_2)$$

As we fixed the values of  $h_1$  and  $h_2$ , the set  $\{g \in G : \sigma(g, h) = a\}$  can be written as:

$$\bigcup_{a_1 + a_2 = a + \sigma(h_1, h_2)} (\{g \in G : \sigma(g, h_1) = a_1 \cdot h_1\} \cap \{g \in G : \sigma(gh_1, h_2) = a_2\})$$

This union is finite since the weak-boundedness guarantees that each term in the union is empty except for finitely many values of  $a_1, a_2$ . Each term is also  $L$ -rational since  $\{g \in G : \sigma(gh_1, h_2) = a_2\} = \{g \in G : \sigma(g, h_2) = a_2\}h_1^{-1}$  is  $L$ -rational by Lemma 1.7.3.

The second statement is an immediate consequence of the first and Lemma 1.7.2.  $\square$

Let  $L$  be a prefix-closed finite-to-one biautomatic structure of  $G$  over the language  $X$ , and  $\sigma \in Z^2(G, A)$  regular. Consider the finite set:

$$Y := \{s(x)i(-\sigma(g, x)) : g \in G, x \in X\}^{\pm 1} \subseteq E$$

For each  $v = x_1 \dots x_n \in X^*$ , we choose  $v' \in Y^*$  such that  $v'(i)_E = s(x_1 \dots x_n)$  for each  $i \geq 1$ . This is possible by the group law on  $E$  in Corollary 1.10.4. Define the language  $L' = \{v' : v \in L\} \subseteq Y^*$ .

**Lemma 1.10.11.**  *$L'$  as described above is a regular language and, for any inverse closed (finite) monoid generating set of  $A$ ,  $(Y, L')$  satisfies the following fellow traveler property:*

*There exists  $k > 0$  such that for any  $v, w \in L', y, z \in Y$  such that  $\pi(v_E) = \pi(yw_E z)$ , then  $v$  and  $yw$   $k$ -fellow travel in  $E$ .*

*Proof.* We follow the proof of [12, Proposition 2.2], verifying more carefully that the FSA we construct accepts exactly  $L'$ .

Let  $M$  be an FSA accepting  $L$ , with state set  $S$ . By the regularity of  $\sigma$ , for all  $a \in A, x \in X$ , the language  $\{w \in Y^* : \sigma(w_G, x) = a\}$  is regular, thus accepted by an FSA  $M_{x,a}$  with state set  $S_{x,a}$ . As  $\sigma(G, X)$  is finite, we set  $S_{x,a} = \emptyset$  for  $a \notin \sigma(G, X)$ , and so only finitely many  $S_{x,a}$  are non-empty.

We now construct an FSA  $N$  accepting  $L'$ : with state set  $S \times \prod S_{x,a}$  together with a failure state  $\otimes$ . A transition  $s(x)i(-a)$  maps the state  $(s, \dots, s_{a',x'}, \dots)$  to  $\otimes$  if  $s_{x,a}$  is not an accept state of  $M_{x,a}$  and maps to  $(xs, \dots, xs_{a',x'}, \dots)$  otherwise. The start state is the tuple  $S_0 = (s_0, \dots, s_{x,a,0}, \dots)$  consisting of the start state from each of the  $S, S_{x,a}$ , and  $(s_0, \dots, s_{a',x'}, \dots)$  is an accept state if  $s$  is an accept state of  $M$ .

We now show that  $L' = L(N)$ . We do this by proving for every  $n \geq 0$  that  $L' \cap Y^n = L(N) \cap Y^n \implies L' \cap Y^{n+1} = L(N) \cap Y^{n+1}$ . Firstly, we notice that  $\epsilon \in L' \cap L(N)$  since  $\epsilon' = \epsilon$ . Suppose we have a word  $w' \in Y^*$  of length  $l(w) = n + 1$  with  $w' = v's(x)i(a), v = x_1 \dots x_n \in L$ , and  $w = x_1 \dots x_n x$ . By the group law, we have:

$$w'_E = v'_E s(x)i(-a) = s(v_G)s(x)i(-a) = s(v_G x)i(-a + \sigma(v_G, x))$$

Therefore,  $w' \in L' \iff \sigma(v_G, x) = a$  and  $w \in L$ . This occurs exactly when  $w \in L$  and  $vs_{x,a,0} \in M_{x,a}$ , i.e., when  $w' \in L(N)$ . As  $L$  is prefix-closed, every  $w \in L \cap Y^{n+1}$  is of the form  $v's(x)i(a), v \in L \cap Y^n$ . Therefore we have  $L' \cap Y^{n+1} = L(N) \cap Y^{n+1}$ , so we conclude by induction that  $L(N) = L'$ .

We verify that  $L'$  satisfies the traveller property for any finite generating set  $Z$  of  $A$ . Let  $d_E$  and  $d_G$  be the word metrics on  $E, G$  with respect to the generating sets  $Y \cup Z, X$  respectively, so that for any  $g, g' \in G, d_E(s(g), s(g')) = d_G(g, g')$  by the construction of  $L'$ . Let  $k$  be the two-sided fellow traveller constant of  $L$  and  $k'$  be given by,

$$k' = \max\{l(w) : w \in \sigma(G, X) \cup \sigma(X, G)\}$$

For  $v', w' \in L'$  and  $y \in Y$  such that  $\pi(v_E) = \pi(yw_E)$ , we have  $\pi(y) \in X$  and  $\pi(w'_E) = w_G$  and similar for  $v$ , so  $d_G(\pi(v(t)_E), \pi(yw(t)_E)) \leq k$  for all  $t \leq 0$  by the fellow travelling of  $L$ . Hence,

$$d_E(s(\pi(yw'(t)_E)), s(\pi(v'(t)_E))) \leq k$$

Now, one obtains  $d_E(s(\pi(yw'(t)_E)), s(\pi(y))w'(t)_E) \leq k'$  and  $d_E(s(\pi(y)), y) \leq k'$  as the differences  $(s(\pi(yw'(t)_E)))^{-1}s(\pi(y))w'(t)_E$  and  $s(\pi(y))y^{-1}$  are in  $\sigma(X, G)$  and  $\sigma(G, X)$  respectively. Since  $s(\pi(v'(t))) = v'(t)$ , the triangle inequality gives us,

$$d_E(yw'(t), v'(t)) \leq k + 2k'$$

so  $yw', v'$   $k + 2k'$ -fellow travel.  $\square$

**Remark 1.10.12.** In [12, Proposition 2.2], there is no requirement that  $L$  is prefix-closed. We believe that this is a necessary assumption for the proof, as otherwise we can find for some  $n, w \in L \cap Y^{n+1}$  not of the form  $v's(x)i(a), v \in L \cap Y^n$ , and thus not accepted by the FSA.

We now are in position to obtain a biautomatic structure of  $E$ :

**Theorem 1.10.13.** For any  $G$ -invariant monoid generating set  $Z$  of  $A$ , with  $Z = Z^{-1}$ . Taking any  $G$ -invariant biautomatic structure  $L_A \subseteq Z^*$  then  $L'L_A$  is a biautomatic structure of  $E$ .

*Proof.*  $L'L_A$  is the concatenation of two regular languages so is also regular. Let  $w_1, w_2 \in L', v_1, v_2 \in L_A, x, y \in Y \cup Z$  such that  $xw_1v_1y =_E w_2v_2$ . Then,  $\pi(xw_1y) =_G \pi(w_2)$ , since  $\pi(v_1) = \pi(v_2) = e$ . By the previous lemma,  $xw_1$  and  $w_2$   $k$ -fellow travel for some  $k \geq 0$ . Therefore,  $d_E(v_1, v_2) \leq k + 1$ , and as  $A \leq E$ , we have  $d_A(v_1, v_2) \leq c$  for some  $c > 0$ . Then  $v_1, v_2$   $ck_A$  fellow-travel where  $k_A$  is the fellow-travel constant of  $A$ .

Since there is a bijection  $L' \rightarrow L$  and the natural map  $L' \rightarrow E \xrightarrow{\pi} G$  is identical to the map  $L' \rightarrow L \rightarrow G$ , we can use [3, Theorem 7.2.8] to obtain a so-called departure function, that is, there is some  $\delta > 0$  such that any sub-word  $u$  of length  $\geq \delta$  of any word of  $L'$  satisfies  $d_G(e, u_G) > 2k$ . The lengths of  $w_1, w_2$  then differ by at most  $\delta$  since  $xw_1, w_2$   $k$ -fellow travel. With all of these bounds together, in a fashion similar to Theorem 1.6.2, we conclude that  $xw_1v_1$  and  $w_2v_2$   $k + \delta + ck_a + c + 1$ -fellow-travel, as required.  $\square$

#### 1.10.4 Conditions for $L$ -Regularity

Given a group extension  $0 \rightarrow A \rightarrow E \rightarrow G \rightarrow 0$ , we would like to simplify the task of checking whether its representing cohomology class in  $H^2(G, A)$  is  $L$ -regular. We state several results, due to [?, Section 2], with that effect.

We recall that an **equivariant** map between two right  $G$ -modules  $A, A'$  is a homomorphism  $f : A \rightarrow A'$  such that for any  $g \in G, a \in A$ ,  $f(a \cdot g) = f(a) \cdot g$ . Such an  $f$  induces

$$f^* : Z^2(G, A) \rightarrow Z^2(G, A'), \sigma(g, h) \mapsto f(\sigma(g, h))$$

for each  $g, h \in G$ . This map furthermore descends to  $f^* : H^2(G, A) \rightarrow H^2(G, A')$ .

**Theorem 1.10.14.** *Let  $f : A \rightarrow A'$  be an equivariant map of right  $G$ -modules with finite  $G$ -action and let  $f$  have finite kernel and cokernel. If  $L$  be a finite-to-one biautomatic structure of  $G$ , then  $[\sigma] \in H^2(G, A)$  is  $L$ -regular if and only if  $[f(\sigma)] = f([\sigma]) \in H^2(G, A')$  is  $L$ -regular.*

Applying this theorem to the map  $A \rightarrow A$  given by multiplying by a non-zero integer yields the important:

**Theorem 1.10.15.** *A cocycle  $\sigma \in H^2(G, A)$  is regular if and only if it is virtually regular, that is, a non-zero multiple of  $\sigma$  is regular.*

**Corollary 1.10.16.** *Let  $F$  be a finite abelian group, then every class in  $H^2(G, F)$  is  $L$ -regular.*

*Proof.* Let  $m$  be the order of  $F$ . As  $\forall x \in F, x^m = e$ , then every cocycle has a multiple that is the cocycle  $\epsilon : G^2 \rightarrow A, (g, h) \mapsto 0$ .  $\square$

Theorem is also used in [?] to prove the following:

**Corollary 1.10.17.** *Let  $H \leq G$  be a subgroup of finite index. Then a cohomology class in  $H^2(G, A)$  is  $L$ -regular if and only if its restriction to  $H^2(H, A)$  is  $L_H$ -regular, where  $L_H$  is the biautomatic structure on  $H$  induced by  $L$ .*

# Chapter 2

## Biautomatic Structures of 3 and 4-Manifolds

In this chapter, we will use the theory of automatic groups that we previously developed to determine biautomatic structures of 3-manifolds.

We assume this time that the reader has a working knowledge of algebraic topology, especially fundamental groups, as well as some familiarity with smooth and Riemannian manifolds. We will not need certain definitions to their full precision, in which case we will defer those elsewhere for the interested reader.

Unless otherwise specified, we will take all manifolds to be compact.

### 2.1 Geometries

In this section, we define what it means for a manifold to have a geometry. We begin with the following lemma:

**Lemma 2.1.1** ([13, Theorem 1.5.4 and Proposition 1.5.8]). *The isometry group of a Riemannian manifold has a canonical Lie Group structure and for any discrete  $G \leq \text{Isom}(M)$ , then  $M/G$  has a natural Riemannian manifold structure such that the covering  $q : M \rightarrow M/G$  is a local isometry.*

Now we have the necessary to define a geometry of a manifold:

**Definition 2.1.2.** *A manifold  $M$  possesses a **geometry modelled on a Riemannian manifold**  $X$  if  $M$  is the quotient of  $X$  by a subgroup  $\Gamma$  of  $\text{Isom}(X)$  such that the natural quotient map is a covering. We also call  $M$  an  $X$ -manifold.*

Importantly, when  $M$  is an  $X$ -manifold, then  $\pi_1(M)$  is a discrete subgroup of  $\text{Isom}(X)$ . We now consider the different maximal geometries that arise. In the 2-dimensional case, the situation is simple: every surface is modelled on either  $S^2$ ,  $\mathbb{R}^2$ , or  $\mathbb{H}^2$ . The situation is more complex in higher dimensions: 3-manifolds need not be geometric, but can be decomposed into pieces modelled on one of 8 geometries. In dimensions 4 and higher, there are manifolds with no such decomposition.

### 2.2 Decomposition of Compact 3-Manifolds

We now describe the decomposition of 3-manifolds into pieces admitting one of 8 geometries. We recall that the **connected sum**  $M \# N$  of 3-manifolds  $M, N$  is

obtained by remove a 3-disc and gluing along the resulting boundary 2-spheres, a precise definition can be found in, for example, [14, Section 2].

**Lemma 2.2.1** ([14, Lemma 2.1]). *The connected sum operation is well defined, associative and commutative up to orientation preserving diffeomorphisms, with identity  $S^3$ .*

The connected sum operator has a rather algebraic flavor. In analogy to rings, the operation yields the concept of a prime manifold<sup>1</sup>.

**Definition 2.2.2** (Prime Manifold). *A manifold  $M$  is a prime manifold if whenever  $M = M_1 \# M_2$ , then  $M_1$  or  $M_2$  is a copy of  $S^3$ .*

Kneser's and Van Kampen's theorems allow us to reduce the problem of automaticity of fundamental groups of compact 3-manifolds to that of prime 3-manifolds. Proofs of Kneser's and Van Kampen's theorems can be found in [13, Theorem 9.2.29] and [15, Page 44] respectively.

**Theorem 2.2.3** (Kneser's Theorem). *Every compact oriented 3-manifold  $M$  has prime decomposition  $M = M_1 \# \dots \# M_n$  for some  $n \in \mathbb{N}$ , unique up to reordering the factors and adding/removing copies of  $S^3$*

**Theorem 2.2.4** (Van Kampen's Theorem). *Let  $X$  be a topological space that is the union of two path-connected open sets  $A$  and  $B$ . Suppose that  $A \cap B \neq \emptyset$  and is path connected. Then, choosing any point  $x_0 \in A \cap B$ , we have:*

$$\pi_1(X) = \pi_1(A) *_{\pi_1(A \cap B, x_0)} \pi_1(B)$$

**Corollary 2.2.5.** *Let  $M$  be a compact oriented 3-manifold with prime decomposition  $M = M_1 \# \dots \# M_n$ , for some  $n \in \mathbb{N}$ . Then*

$$\pi_1(M) = \pi_1(M_1) * \dots * \pi_1(M_n)$$

*Proof.* We suppose that  $n = 2$ , as an obvious induction yields the general result.

Consider the open covers  $M = A \cup U$ ,  $N = B \cup V$  where  $U, V$  are the interiors of the disk  $D^3$  that we remove from  $M$  and  $N$  to perform the connected sum and  $A, B$  are  $M, N$  without a point in  $U, V$  respectively. Then we have an open cover  $M \# N = X \cup Y$  where  $M \setminus U \subset X$ ,  $N \setminus V \subset Y$  and  $X, Y$  are homotopy equivalent to  $A, B$  respectively.

Finding such  $X$  and  $Y$  is always possible, since in the worst case we may choose to perform the connected sum operation with embedded disks  $U', V'$  with closures homeomorphic with  $D^3$  contained entirely in  $U, V$  respectively, thereby allowing us to have  $X$  and  $Y$  disjoint from  $U'$  and  $V'$  yet  $X \cap N \subset V$  and  $Y \cap M \subset U$ .

Now, we apply Van Kampen's theorem to  $M, N$ , and  $M \# N$ . Firstly,

$$\pi_1(A \cap U) = \pi_1(B \cap U) = \pi_1(X \cap Y)$$

because  $\pi_1(S^2) = 0$  and these intersections are neighborhoods of the two glued boundaries of the embedded  $D^3$  of  $M$  and  $N$ . Secondly,  $\pi_1(U) = \pi_1(V) = 0$  because

---

<sup>1</sup>Perhaps confusingly, the analogy in question is between prime manifolds and *irreducible* elements of a ring.



they are contractible and  $\pi_1(A) = \pi_1(X)$ ,  $\pi_1(B) = \pi_1(Y)$  due to their homotopy equivalence. Hence,

$$\pi_1(M) = \pi_1(A) * \pi_1(U) = \pi_1(X) \text{ and } \pi_1(N) = \pi_1(B) * \pi_1(V) = \pi_1(Y)$$

A final use of Van Kampen yields:

$$\pi_1(M \# N) = \pi_1(X) * \pi_1(Y) = \pi_1(M) * \pi_1(N)$$

as required.  $\square$

With this and our work in the previous chapter, we obtain:

**Corollary 2.2.6.** *A 3-manifold  $M$ 's fundamental group is (bi)automatic structure if and only if the fundamental groups of the manifolds in its prime decomposition are (bi)automatic.*

$M$ 's prime components can then be decomposed into geometric pieces.

**Theorem 2.2.7** (Thurston's Geometrization Conjecture). *Every oriented, closed, prime 3-manifold can be cut along tori so that the interiors of the resulting manifolds admit one of the following geometries:*

$$S^3, \mathbb{R}^3, \mathbb{H}^3, S^2 \times \mathbb{R}, \mathbb{H}^2 \times \mathbb{R}, \widetilde{\text{SL}}_2, \text{Nil}^3, \text{Sol}^3$$

In [3, Chapter 12], it is proved that compact 3-manifolds have automatic fundamental groups if none of the manifolds resulting from the decompositions above are modelled on  $\text{Nil}^3$  nor  $\text{Sol}^3$ . Here, we discuss biautomaticity when the decomposition above is trivial<sup>2</sup>, as well as introducing further necessary geometric notions required both here and in the arguments of [3, Chapter 12].

## 2.3 2-Dimensional Orbifolds

Informally, a **2-orbifold** is a second-countable topological manifold where every point has a neighborhood isomorphic to  $\mathbb{R}^2/\Gamma$  with  $\Gamma$  a finite subgroup of  $\text{Isom}(\mathbb{R}^n)$ . Points for which  $\Gamma$  is non-trivial are called **singular**. A formal definition can be found in [16, Page 422].

**Theorem 2.3.1** ([17, Theorem 1]). *The singular points of a 2-dimensional orbifold consists are of 3 types:*

- *mirror points: homeomorphic to  $\mathbb{R}^2/Z_2$ , where  $Z_2$  acts by reflection along a line.*
- *cone points: homeomorphic to  $\mathbb{R}^2/Z_n$ , where  $Z_n$  is generated by rotation of  $2\pi/n$ .*
- *corners reflectors: homeomorphic to  $\mathbb{R}^2/D_n$ , where  $D_n$  is the dihedral group of order  $2n$ , that is generated by reflection about two lines that meet at an angle of  $\pi/n$ .*

---

<sup>2</sup>Biautomaticity of the fundamental group is in general an open problem when the decomposition is non-trivial, including for graph manifolds, due to the amalgamations by  $\mathbb{Z}^2$  that arise.

Given a surface  $X$ , we denote by  $X(\alpha_1, \dots, \alpha_n)$  an orbifold with isolated cone points of order  $\alpha_i$ , homeomorphic to  $X$ . Many topological definitions can be extended to the category of orbifolds, for example:

**Definition 2.3.2.** A continuous map  $p : X \rightarrow Y$  between two orbifolds is called an **orbifold covering map** if it satisfies the following condition for every  $y \in Y$ , there is a neighborhood  $U_y \ni y$  and a discrete space  $D$  such that  $p^{-1}(U_y) = \bigsqcup_{d \in D} V_d$  where for each  $d \in D$ ,  $p|_{V_d} : V_d \rightarrow U_y$  is a natural quotient map between two quotients of  $\mathbb{R}^2$  by finite groups, one contained in the other. In other words, if  $U_y \cong \mathbb{R}^2/\Gamma$ , then each  $V_d \cong \mathbb{R}^2/H$  for some  $H \leq \Gamma$  and  $p|_{V_d}$  is the natural quotient map.

This definition is nearly identical to that of a (topological) covering map, except that the  $p|_{V_d}$  aren't in general homeomorphisms.

**Definition 2.3.3.** Let  $Y$  be an orbifold. Then:

- The **universal orbifold cover** of  $Y$  is an orbifold  $E$ , with orbifold covering map  $f : E \rightarrow Y$  such that for any other orbifold covering map  $p : X \rightarrow Y$  we have an orbifold covering map  $g : E \rightarrow X$  such that  $f = p \circ g$ .
- $Y$  is **good** if its universal cover is a manifold, and **bad** otherwise
- The **orbifold fundamental group** of  $Y$ , denoted  $\pi_1(Y)$ , is the group of orbifold deck transformations of its universal orbifold cover.

If  $X$  is an orbifold, we always take  $\pi_1(X)$  to mean the orbifold fundamental group, not that of the underlying topological manifold. The following results show us that the orbifold fundamental group always exists, and furthermore gives us a method to calculate orbifold fundamental groups without corner reflectors.

**Lemma 2.3.4.** The universal orbifold covering always exists.

*Proof.* This is true for any dimension, but we only prove it for 2-orbifolds, following the argument in [16, page 423]. Let  $M'$  be a 2-orbifold without boundary. If the underlying surface has boundary, then take the double of  $M'$ , which we denote  $M$ , is a 2-fold orbifold covering of  $M'$  with no boundary, and furthermore has only cone points as singularities.  $M'$  and  $M$  share the same universal cover, if it exists, hence we only need to find a universal cover for  $M$ .

Let  $\Sigma$  be the union of  $M$ 's cone points. By the definition of an orbifold covering, an orbifold covering map  $p : X \rightarrow M$  restricts to a topological covering  $X - p^{-1}(\Sigma) \rightarrow M - \Sigma$ . Moreover, for any cone point  $x \in \Sigma$  of angle  $2\pi/n$ , then any cone point in  $p^{-1}(x)$  has angle  $2\pi/m$  where  $m$  divides  $n$ .

Let  $N$  be the orbifold  $M$  with small cones centred at the cone points removed, and  $Y = p^{-1}(N)$ . Then both  $Y$  and  $N$  are 2-manifolds, and the orbifold covering  $p$  restricts to a topological covering  $Y \rightarrow N$ . Consider a circle  $C$  of  $\partial N$ , winding once around a cone of angle  $2\pi/n$ , and  $\alpha$  a representative of  $C$  in  $\pi_1(N)$ . By our earlier remark about cone angles, the circles that form the pre-image of  $C$  wind around their respective cone points by a natural number dividing  $n$ . Therefore,  $p_1(Y)$  contains all conjugates of  $\alpha^n$ .

We now enumerate the cone points of  $\Sigma$  as  $x_1, \dots, x_t$ , cone angles  $2\pi/n_i$  respectively, and denote the respective circle in  $\partial N$  as  $C_i$  and its representative in  $\pi_1(N)$  as

$\alpha_i$ . We obtain a group  $G$  from  $\pi_1(N)$  by adding the relations  $\alpha_i^{n_i}$  to its presentation. Then the kernel  $K$  of the natural (group) quotient map  $\pi(N) \rightarrow G$  is contained in  $\pi_1(Y)$ . By the Galois correspondence of topological covers, there is a topological cover  $Z$  of  $N$  corresponding to  $K$ . By adding appropriate cones to  $Z$ , we obtain an orbifold covering of  $M$ . As  $K$  is the smallest possible subgroup of  $\pi_1(N)$  that a topological covering of  $N$  extending to an orbifold covering  $Z'$  of  $M$  can have, then  $Z$  is universal with respect to these coverings. Therefore,  $Z'$  is the universal orbifold cover of  $M$ .  $\square$

We remark that the group  $G$  in the proof above is the orbifold fundamental group of  $M$ . Hence:

**Corollary 2.3.5.** *Let  $X = \Sigma(p_1, \dots, p_m)$  be a 2-orbifold with no reflector curves. Let  $N$  be the surface obtained from  $\Sigma$  by removing a neighborhood of the cone points removed. Let  $\alpha_i$  be representatives in  $\pi_1(N)$  of the circle in  $\partial N$  around the cone point  $x_i$ , respectively. Then  $\pi_1(X)$  is the quotient of  $\pi_1(N)$  obtained by adding the relations  $\alpha_i^{p_i} = e$ .*

Definition 2.1.2 also applies to (good) orbifolds. Like with surfaces, every good 2-orbifold is modelled on  $S^2$ ,  $\mathbb{R}^2$ , or  $\mathbb{H}^2$ , determined by their Euler characteristic. See [16, Pages 425-427] for more details.

## 2.4 Seifert Fibre Spaces

Seifert fibre spaces correspond to 6 of the 3-dimensional geometries we consider, by Theorem 12.1.1. in [13]:

**Theorem 2.4.1.** *A closed orientable 3-manifold has a geometric structure modelled on one of the following 6 geometries:*

$$S^3, \mathbb{R}^3, S^2 \times \mathbb{R}, \mathbb{H}^2 \times \mathbb{R}, \widetilde{\text{SL}}_2, \text{Nil}$$

*if and only if it is a Seifert manifold.*

Seifert fibre spaces are 3-manifolds that expressed as a disjoint union of circles, that we call fibres. We define them as in [13]; a slight generalization of Seifert's original definition.

**Definition 2.4.2** (Seifert Fibre Spaces). *We define Seifert fibre spaces in 4 steps:*

- A **trivial fibred solid torus** is simply a torus  $D^2 \times S^1$  with the foliation<sup>3</sup> by the circles  $p \times S^1$ , for  $p \in D^2$ .
- A **fibred solid torus** is a solid torus with a foliation by circles that is finitely covered by a trivial fibred torus.
- Similarly, a **fibred solid Klein bottle** is a solid Klein bottle that is finitely covered by a trivial fibred torus.

---

<sup>3</sup>A foliation is a decomposition of a manifold into a union of disjoint submanifolds that satisfy a particular property. See, for example, [18]

- Finally, a **Seifert fibre space** is a 3-manifold  $M$  that can be expressed as a disjoint union of circles, called fibres, such that each fibre has a neighborhood in  $M$  consisting of fibres that is isomorphic to a fibred solid torus or Klein bottle. The fibres form a foliation of  $M$ .

**Definition 2.4.3.** Let  $M$  be a Seifert fibre space. A fibre in  $M$  is **regular** if it has a neighborhood isomorphic to a trivial fibred torus, and critical otherwise.

Now, consider the space  $X$  obtained from the (topological) quotient of  $M$  by identifying any two points of a single fibre together.  $X$  then has the structure of a 2-orbifold, and we call  $X$  the base orbifold of  $M$ . The singular points  $X$  correspond to the critical fibres; cone points correspond to orientation preserving critical fibres while mirror points correspond to orientation reversing critical fibres.  $X$  furthermore has no corner reflectors.

The previous paragraph and the following two lemmas are due to [13, Pages 430-431].

**Lemma 2.4.4.** The universal covering,  $\tilde{M}$ , of a Seifert fibre space  $M$ , is either also foliated by circles, and again is a Seifert fibre space, or is foliated by lines. In either case, there is an induced orbifold covering from  $\tilde{M}$ 's base orbifold to  $M$ 's base orbifold.

**Lemma 2.4.5.** Let  $M$  be a Seifert fibre space without boundary. Then its universal cover,  $\tilde{M}$ , is homeomorphic to one of  $S^3, \mathbb{R}^3$ , or  $S^2 \times \mathbb{R}$ . Furthermore, the induced foliation of  $\tilde{M}$  by circles or lines is of structure:

- a Seifert bundle over one of the orbifolds  $S^2, S^2(p), S^2(p, q)$ , where  $p, q$  are coprime.
- a product line bundle over  $\mathbb{R}^2$
- a product line bundle over  $S^2$

The usefulness of Seifert fibred spaces, for our purposes, is due to the following group extension, [16, Lemma 3.2] and [13, Proposition 10.4.2].

**Theorem 2.4.6.** Let  $M$  be a Seifert fibre space with base orbifold  $X$ , then we have an exact sequence:

$$e \rightarrow C \rightarrow \pi_1(M) \rightarrow \pi_1(X) \rightarrow e$$

where  $C \leq \pi_1(M)$  is cyclic and generated by a regular fibre.  $C$  is only finite if  $M$  is covered by  $S^3$ .

**Remark 2.4.7.** By "generated by a regular fibre", we mean that  $C$  is generated by a representative of the regular fibre in  $\pi_1(M)$ . Due to  $C$  being normal in  $\pi_1(M)$ , a base point need not be specified.

*Proof.*  $\pi_1(M)$  acts on its universal cover  $\tilde{M}$  by deck transformations that preserve the foliation. Therefore, there is an induced action, by deck transformations, of  $\pi_1(M)$  on  $\tilde{X}$ , the base orbifold of  $\tilde{M}$ . This induces an action of  $\pi_1(M)$  on  $\pi_1(X)$ , and thus a surjective homomorphism  $\pi_1(M) \rightarrow \pi_1(X)$ .

The kernel of this homomorphism, which we call  $C$ , must be covering translations of  $\tilde{M}$  projecting to the identity on  $\tilde{X}$ .  $C$  acts freely on any of the fibres of  $\tilde{M}$ , which is infinite cyclic when  $\tilde{M}$  is not compact. By the previous lemma, this occurs exactly when  $M$  is not covered by  $S^3$ .  $\square$

## 2.5 The Isotropic Geometries

In this section, we show that every manifold modelled on  $\mathbb{R}^n$ ,  $S^n$ , or  $\mathbb{H}^n$ , for any  $n \in \mathbb{N}$  has biautomatic fundamental group. As every 2-dimensional manifold is modelled on one of the geometries,  $\mathbb{R}^2$ ,  $\mathbb{H}^2$ ,  $S^2$ , we see that the fundamental group of any 2-dimensional manifold is biautomatic.

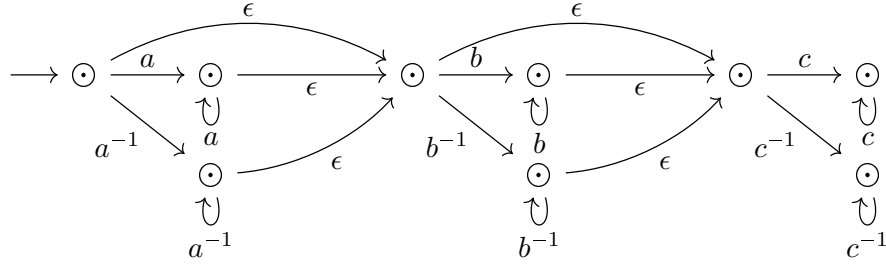
### 2.5.1 The Geometry of $\mathbb{R}^n$

$\mathbb{R}^n$ -manifolds are often referred as **flat** manifolds. Every  $\mathbb{R}^n$ -manifold has fundamental group virtually  $\mathbb{Z}^n$ , which we know to be biautomatic.

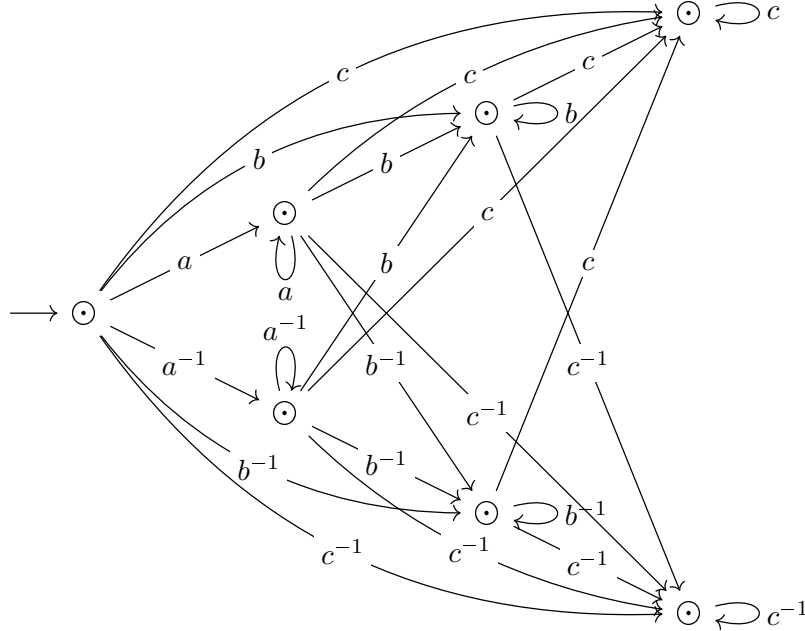
We now provide some examples, beginning with the 3-torus. We let  $\mathbb{Z}^3$  be generated by  $a, b, c$ . Recalling the biautomatic structure for  $\mathbb{Z}$  in Lemma 1.8.4, then  $(\{a, b, c\}, L)$  is a biautomatic structure of  $\mathbb{Z}^3$ , where:

$$L = (\{a\}^* \cup \{a^{-1}\}^*)(\{b\}^* \cup \{b^{-1}\}^*)(\{c\}^* \cup \{c^{-1}\}^*)$$

$L$  is accepted by this NFA:



We obtain from this NFA a DFA accepting  $L$  by deleting the states that are end-points of  $\epsilon$  transitions and suitably replacing affected transitions.



As  $L^{-1} = (\{c\}^* \cup \{c^{-1}\}^*)(\{b\}^* \cup \{b^{-1}\}^*)(\{a\}^* \cup \{a^{-1}\}^*)$ , swapping  $a$  and  $c$  in the FSA above yields FSA accepting  $L^{-1}$ . We remark that there is an obvious generalization for  $n$ -tori for  $n \in \mathbb{N}$ .

The Hantzsche–Wendt manifold, studied in detail in [19], has fundamental group given by:

$$G := \langle x, y, z | xy^2x^{-1}y^2 = yx^2y^{-1}x^2 = xyz = e \rangle$$

We denote by  $\Lambda$  the normal subgroup  $\langle x^2, y^2, z^2 \rangle \cong \mathbb{Z}^3$  and we have  $G/\Lambda \cong \mathbb{Z}_2^2$ , with a set of coset representatives  $P = \{e, x, y, z\}$ . We now construct a biautomatic structure for  $G$ .

We note the relations  $xz^2x^{-1}z^2 = yz^2y^{-1}z^2 = zx^2z^{-1}x^2 = zy^2z^{-1}y^2 = e$  that are readily obtained from the ones in the presentation. The action by conjugation of elements of  $P' = P \setminus \{e\}$  as in Theorem 1.10.7 is then  $g \rightarrow g^{-1}$  for  $g \in \{x^2, y^2, z^2\}^{\pm 1}$ . Taking  $a = x^2, b = y^2, c = z^2$  and  $L$  as before, we see that  $L$  is invariant under this action, since words in  $L$  are of form  $a^pb^qc^r$ ,  $p, q, r \in \mathbb{Z}$ , with  $a^{-p}b^{-q}c^{-r}$  clearly in  $L$ . Therefore,  $(\{x^2, y^2, z^2\}^{\pm 1} \cup P, L \cup LP')$  is a biautomatic structure of  $G$ .  $P'$  is accepted by the DFA:

$$\longrightarrow \bullet \begin{array}{c} \xrightarrow{x} \\ \xrightarrow{y} \\ \xrightarrow{z} \end{array} \odot$$

Hence  $LP$  is accepted by the NFA and DFA above but with adding transitions labelled  $x, y, z$  from every accept state to a new accept state.

## 2.5.2 The Geometry of $\mathbb{H}^n$

The matter of biautomaticity for this geometry is easily settled by following theorem [3, Theorem 3.4.1].

**Theorem 2.5.1.** *The fundamental group of a negatively curved compact manifold or orbifold or convex manifold with boundary is hyperbolic. In particular, hyperbolic compact manifolds and orbifolds have hyperbolic fundamental group, and are thus strongly geodesically automatic.*

A family of standard examples are the compact oriented surfaces of genus  $g \geq 2$ , which have fundamental group:

$$\pi = \langle a_1, b_1, \dots, a_g, b_g | [a_1, b_1] \dots [a_g, b_g] \rangle$$

We describe an automaton, constructed as in Theorem 1.9.1, over  $A := \{a_i, b_i | i = 1, \dots, g\}^{\pm 1}$  accepting the geodesic words with respect to this presentation:

- Let the state set  $S$  be the set of reduced words of length  $\leq 2g$  with start state the empty word.
- For every  $w \in S$ ,  $a \in A$ , if  $wa \in S$ , then there is an  $a$  transition  $w \rightarrow aw$  labelled  $a$ .
- If  $w' \in W$  of length  $2g$  can be obtained from  $w$  by deleting its first letter and adding  $a \in A$  at the end, then there is an  $a$  transition  $w \rightarrow w'$ , unless  $wa$  is the first  $2g+1$  letters of the relation of  $\pi$ , in which case the transition's target is the failure state instead.
- all remaining arrows have target the failure state.

### 2.5.3 The Geometry of $S^n$

All manifolds modelled on  $S^n$  for some  $n$  have finite fundamental group by Bonnet's theorem [20, page 200], yielding biautomatic structures that are easily described given a list of all its elements. In the case  $n = 3$ , we furthermore have that a 3-manifold has finite fundamental group if and only if it is an  $S^3$ -manifold, by Theorem 2.4.6.

We take as examples the Lens space  $L(p, q)$ , where  $p, q$  are coprime, as examples. They are quotients of  $S^3$  by a  $\mathbb{Z}/p$  action and have fundamental group  $\mathbb{Z}/p \cong \langle x | x^p \rangle$ . Clearly,  $(\{x, x^{-1}\}, \{\epsilon, x, xx, xxx, \dots, xx\dots x \text{ (} p-1 \text{ times)}\})$  is a biautomatic structure accepted by FSA:

$$\longrightarrow \odot \xrightarrow{x} \odot \xrightarrow{x} \odot \xrightarrow{x} \dots \xrightarrow{x} \odot$$

where there are  $p - 1$  accept states.

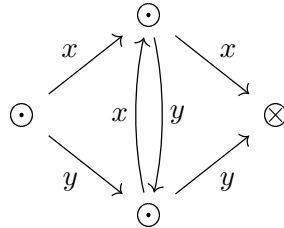
## 2.6 Fundamental Groups of the 8 Geometries

We describe biautomatic structures for the remaining 3-dimensional geometries, except  $\text{Nil}^3$ - and  $\text{Sol}^3$ -manifolds, as their fundamental groups are not even asynchronously automatic, see Section 2.8.

### 2.6.1 The Geometry of $S^2 \times \mathbb{R}$

There are only 7 manifolds with  $S^2 \times \mathbb{R}$  geometry [16, page 457], with fundamental groups either  $\mathbb{Z}$  or the infinite dihedral group  $D = \langle x, y | x^2 = y^2 = e \rangle$  [21, page 16]. As we already know,  $\mathbb{Z}$  is biautomatic, hence we need only to verify that  $D$  is biautomatic:

Let  $A = \{x, y\}$  and  $L$  be the language accepted by the FSA:



so that  $w \in L$  is a reduced word of form with entries alternating between  $x$  and  $y$ . Clearly, every element of  $D$  has a representative in  $L$ , and any  $v, w \in L, d(v, w) \leq 1$  then they are equal or one is a prefix of the other, so they 1-fellow travel. Since  $L^{-1} = L$ ,  $(A, L)$  is a biautomatic structure of  $D$ .

### 2.6.2 The Geometries of $\mathbb{H}^2 \times \mathbb{R}$ and $\widetilde{\text{SL}}_2$

We can show that manifolds modelled on either of these two geometries have bi-automatic fundamental group with the same method, using the fact that the base orbifold is hyperbolic. The base's orbifold fundamental group is then a discrete subgroup of  $\text{Isom}(\mathbb{H}^2)$ . We now have the following theorem, due to [12].

**Theorem 2.6.1.** *Let  $G$  be a discrete finitely-generated subgroup of  $\text{Isom}(\mathbb{H}^2)$  and  $A$  finitely-generated abelian. For any virtually central extension  $0 \rightarrow A \xrightarrow{i} E \xrightarrow{\pi} G \rightarrow 0$ ,  $E$  is biautomatic.*

*Proof.* It suffices to prove that every class in  $H^2(G, A)$  is  $L$ -regular,  $L = \mathcal{G}(G, X)$  for some generating set  $X$ . The  $L$ -regularity of  $[\sigma] \in H^2(G, A)$  is guaranteed by the  $L_H$ -regularity of the restriction of  $[\sigma]$  to  $H^2(H, A)$ , where  $H$  is a finite-index subgroup of  $G$ , and  $L_H$  is the biautomatic structure of  $H$  induced by  $G$ . We can, in particular, choose  $H$  such that it is torsion-free and acts trivially on  $A$ . As  $H$  is a subgroup of finite-index, it is also hyperbolic. Therefore  $S\mathcal{U}(H)$  is a singleton, so  $L_H \sim \mathcal{G}(H, Y)$  for some generating set  $Y$ . Without loss of generality, we can suppose that  $G$  is torsion-free and acts trivially on  $A$ , thanks to Lemma 1.10.10.

As  $A$  is finitely-generated abelian, it can be split into the direct sum of a finite group and finitely many copies of  $\mathbb{Z}$ . Hence,

$$H^2(G, A) = H^2(G, F) \oplus \bigoplus_{i=1}^n H^2(G, \mathbb{Z})$$

As every element of  $H^2(G, F)$  is  $L$ -regular, the problem is then reduced to constructing a regular cocycle for a non-zero multiple of the generator of  $H^2(G, \mathbb{Z})$ . Such a construction using hyperbolic geometry is provided in [?, Section 3].  $\square$

**Corollary 2.6.2.** *Fundamental groups of manifolds with geometries modelled on  $\mathbb{H}^2 \times \mathbb{R}$  and  $\widetilde{\text{SL}}_2$  are biautomatic*

*Proof.* Such a manifold  $M$  is Seifert-fibred with exact sequence  $e \rightarrow \mathbb{Z} \rightarrow \pi_1(M) \rightarrow \pi_1(X) \rightarrow e$ , where  $X$  is a hyperbolic 2-orbifold, so that  $\pi_1(M) \leq \text{Isom}(\mathbb{H}^2)$ . As  $\text{Aut}(\mathbb{Z})$  is finite, then the extension is virtually central, and the previous theorem applies.  $\square$

Taking  $M$  as in the proof above, we furthermore obtain, by Theorem 1.10.13, that an automatic structure of  $\pi_1(M)$  is given by  $L'L_{\mathbb{Z}}$  where  $L'$  is obtained as for the proof above Theorem 1.10.13 from  $L = \mathcal{G}(\pi_1(X), B)$  for any generating set  $B$ .  $L_{\mathbb{Z}}$  is given as in the proof of Lemma 1.8.4. This choice of biautomatic structure of  $\mathbb{Z}$  is always suitable since we proved that it is invariant under inversion, which is the only possible non-trivial action induced by the extension.

We now take as examples the unit tangent bundle  $U$  of a genus  $g \geq 2$  surface  $\Sigma$ . These are  $\widetilde{\text{SL}}_2$ -manifolds with exact sequence:

$$0 \rightarrow \mathbb{Z} \rightarrow \pi_1(U) \xrightarrow{p} \pi_1(\Sigma) \rightarrow e$$

Using the Euler number of circle bundle, the fundamental group can be worked out to be [22],

$$\pi_1(U) = \langle a_1, b_1, \dots, a_g, b_g, t | [a_1, b_1] \dots [a_g, b_g] t^{2g-2}, [a_i, t], [b_i, t], 1 \leq i \leq g \rangle$$

where  $t$  is the generator of  $\mathbb{Z}$ . We notice that the induced action on  $\mathbb{Z}$  is trivial since  $t$  commutes with every generator, and that the map  $p$  is then simply given by  $t \mapsto e, a_i \mapsto a_i, b_i \mapsto b_i, 1 \leq i \leq g$ . We use the natural choice for the section  $s : \pi_1(\Sigma) \rightarrow \pi_1(U), a_i \mapsto a_i, b_i \mapsto b_i$ , so that its associated cocycle has image 0 in  $\mathbb{Z}$ . Letting  $X = \{a_1, b_1, \dots, a_g, b_g\}$ , the language  $L'$  obtained from  $L = \mathcal{G}(\pi_1(\Sigma), X)$  is again  $L$ .  $\mathcal{G}(\pi_1(\Sigma), X)L_{\mathbb{Z}}$  is thus a biautomatic structure of  $\pi_1(U)$ .



## 2.7 The 4-Dimensional Geometries

We now discuss the biautomaticity of the fundamental groups of the 4-dimensional geometries. There are 19 classes of maximal 4-dimensional geometries, which we split into 5 types, as found in [23]. We exclude the tangent space of  $\mathbb{H}^2$  as no manifold modelled on it is closed.

| Type                     | Geometry                                                                                                                                                                                   |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Solvable                 | $\mathbb{R}^4, \text{Nil}^3 \times \mathbb{R}, \text{Nil}^4, \text{Sol}_0^4, \text{Sol}_1^4, \text{Sol}^3 \times \mathbb{R}$                                                               |
| Mixed                    | $\mathbb{H}^2 \times S^2, S^3 \times \mathbb{R}, S^2 \times \mathbb{R}^2, \mathbb{H}^3 \times \mathbb{R}, \mathbb{H}^2 \times \mathbb{R}^2$<br>$\widetilde{\text{SL}}_2 \times \mathbb{R}$ |
| Hyperbolic               | $\mathbb{H}^4, \mathbb{H}_{\mathbb{C}}^2, \mathbb{H}^2 \times \mathbb{H}^2$                                                                                                                |
| Finite Fundamental Group | $S^4, \mathbb{CP}^2, S^2 \times S^2$                                                                                                                                                       |

In the literature, the  $\text{Sol}^3 \times \mathbb{R}$  geometry is often instead referred by the infinite family of geometries  $\text{Sol}_{m,n}^4, m, n \in \mathbb{N}$ . These are, in fact, the  $\text{Sol}^3 \times \mathbb{R}$ -manifolds that are Seifert. We defer the case of the solvable geometries to Section 2.8.

### 2.7.1 Seifert Manifolds

We generalize the definition of of Seifert Fibred 3-manifold to any dimension  $n \geq 3$ :

**Definition 2.7.1.** *A Seifert manifold is the total space of an orbifold bundle over a 2-orbifold with compact flat fibres.*

We denote a general fibre of a compact Seifert  $n$ -manifold to be the fibre of a regular point, which is orientable if the Seifert manifold is orientable. Theorem 2.4.6 has a generalization for orbifold bundles, given in [24]. We specialize this generalization to the Seifert manifold case.

**Theorem 2.7.2.** *Let  $M$  be a Seifert  $n$ -manifold with base orbifold  $B$  and general fibre  $F$ . Then we have an exact sequence:*

$$\pi_1(F) \rightarrow \pi_1(M) \rightarrow \pi_1(B) \rightarrow e$$

*The map  $\pi_1(F) \rightarrow \pi_1(M)$  is furthermore injective if  $B$  is good and aspherical.*

There are 8 classes of 4-geometries that are Seifert [25, Page], listed on the table below. We note that, unlike in dimension 3, it is possible that a Seifert 4-manifold does not have a geometric structure.

| Base Orbifold Type | Geometry                                                                                         |
|--------------------|--------------------------------------------------------------------------------------------------|
| Spherical or Bad   | $S^3 \times \mathbb{R}, S^2 \times \mathbb{R}^2$                                                 |
| Flat               | $\mathbb{R}^4, \text{Nil}^3 \times \mathbb{R}, \text{Nil}^4, \text{Sol}^3 \times \mathbb{R}$     |
| Hyperbolic         | $\mathbb{H}^2 \times \mathbb{R}^2, \widetilde{\text{SL}}_2 \times \mathbb{R}$ , or non-geometric |

Except for two  $\mathbb{R}^4$ -manifolds that are not Seifert.

We remind the reader that a manifold  $M$  is aspherical if  $\pi_m(M) = 0$  for  $m \geq 2$ . Hyperbolic manifolds and orbifolds, in particular, satisfy this condition, due to being covered by  $\mathbb{H}^n$ .

**Corollary 2.7.3** ([23, Page 7]). *Let  $M$  be a Seifert 4-manifold with a hyperbolic base orbifold  $B$ . Then we have a short exact sequence:*

$$e \rightarrow \mathbb{Z}^2 \rightarrow \pi_1(M) \rightarrow \pi_1(B) \rightarrow e$$

We furthermore remark that the induced action of  $\pi_1(B)$  on  $\mathbb{Z}^2$  is contained in a dihedral group of order 4 or 6. As a result, the extension is virtually central, thus Theorem 2.6.1 applies.

**Corollary 2.7.4.** *The fundamental group of a Seifert 4-manifold is biautomatic when the base orbifold is hyperbolic. Therefore  $\mathbb{H}^2 \times \mathbb{R}^2$ - and  $\widetilde{\mathrm{SL}}_2 \times \mathbb{R}$ -manifolds have biautomatic fundamental group.*

## 2.7.2 The Non-Seifert Mixed Geometries

By [26, Theorem 10.13],  $S^2 \times \mathbb{R}^2$ -manifolds have fundamental group virtually  $\mathbb{Z}^2$ , hence are biautomatic. In the case of  $S^3 \times \mathbb{R}$ -manifolds, it is proved in [26, Chapter 11] that the fundamental group has two ends, so is virtually  $\mathbb{Z}$  and biautomatic.

The situation is trickier for the geometries  $S^2 \times \mathbb{H}^2$  and  $\mathbb{H}^3 \times \mathbb{R}$ ; the theory we developed alone only allows us to conclude that the fundamental groups are automatic. We now refer to  $PD_2$  groups, defined in [26, Section 1.8]. They are isomorphic to the fundamental group of an aspherical closed 2-manifold, which we have shown to be biautomatic.

By [26, Theorem 10.1],  $S^2 \times \mathbb{H}^2$ -manifolds are virtually  $PD_2$ , so are automatic, and satisfy an exact sequence  $e \rightarrow P \rightarrow \pi \rightarrow C_2 \rightarrow e$  where  $P$  is  $PD_2$ ,  $\pi$  is the fundamental group, and  $C_2$  is the group of order 2. By [27],  $\pi$  is hyperbolic if and only if it doesn't contain a subgroup  $\cong \mathbb{Z}^2$ . This is clearly always the case, so  $\pi$  is strongly geodesically automatic.

Every  $\mathbb{H}^3 \times \mathbb{R}$ -manifold  $M$  has a finite cover diffeomorphic to  $N \times S_1$ , where  $N$  is an  $\mathbb{H}^3$ -manifold [26, Page 182]. Hence,  $\pi_1(M)$  is virtually a product  $H \times \mathbb{Z}$  where  $H$  is a hyperbolic group, thus  $\pi_1(M)$  is automatic.

## 2.7.3 The Hyperbolic Geometries

$\mathbb{H}_\mathbb{C}^2$ -manifolds are strongly geodesically automatic by virtue of Theorem 2.5.1, as in the case of  $\mathbb{H}^4$ .

We shall say that a  $\mathbb{H}^2 \times \mathbb{H}^2$ -manifold is reducible if its fundamental group contains a normal finite-index subgroup isomorphic to a product of  $PD_2$  groups. In this case, we know that the fundamental group is automatic. It is an open problem whether irreducible  $\mathbb{H}^2 \times \mathbb{H}^2$ -manifold fundamental groups are biautomatic.

## 2.8 The Non-Automatic Geometries

We finish our survey of the 3 and 4-dimensional geometries by briefly discussing why the manifolds modelled on

$$\mathrm{Sol}^3, \mathrm{Nil}^3, \mathrm{Sol}^3 \times \mathbb{R}, \mathrm{Nil}^3 \times \mathbb{R}, \mathrm{Nil}^4, \mathrm{Sol}_0^4, \mathrm{Sol}_1^4$$

have fundamental groups not asynchronously automatic, using the results about their isomorphism classes found in [26, Chapter 8] and [13]. We begin with the following theorem:

**Theorem 2.8.1** ([3, Page 169]). *Any asynchronously automatic virtually nilpotent group is virtually abelian.*

Manifolds modelled on  $\text{Nil}^3$ ,  $\text{Nil}^3 \times \mathbb{R}$ ,  $\text{Nil}^4$  are virtually nilpotent but not virtually abelian fundamental groups so are not asynchronously automatic.  $\text{Sol}_1^4$ -manifold fundamental groups are semidirect products of form  $\Gamma \rtimes \mathbb{Z}$ , where  $\Gamma$  is a  $\text{Nil}^3$ -manifold fundamental group. By Theorem 1.8.9,  $\Gamma \rtimes \mathbb{Z}$  being automatic would imply that  $\Gamma$  is automatic, which is not the case. Hence,  $\Gamma \rtimes \mathbb{Z}$  is not automatic. For the remaining geometries, we use the following:

**Theorem 2.8.2** ([28, Page 3]). *Let  $G = \mathbb{Z}^m \rtimes_B \mathbb{Z}$  where  $B \in \text{GL}_m(\mathbb{Z})$  is of infinite order. Then  $G$  is not asynchronously automatic.*

We recall that an element of  $\text{GL}_m(\mathbb{Z})$  of finite order must have eigenvalues that are roots of unity.

$\text{Sol}^3$ - and  $\text{Sol}^3 \times \mathbb{R}$ -manifolds have fundamental group  $\mathbb{Z}^2 \rtimes_B \mathbb{Z}$  and  $\mathbb{Z}^3 \rtimes_B \mathbb{Z}$  respectively. In the  $\text{Sol}^3$  case,  $B$  has real eigenvalues not equal to  $\pm 1$ , so it's impossible for the eigenvalues to be roots of unity, hence the non-automaticity. In the  $\text{Sol}^3 \times \mathbb{R}$  case, all eigenvalues are real and distinct, so at least one is not  $\pm 1$ , so not a root of unity. Finally,  $\text{Sol}_1^4$ -manifolds have fundamental group  $\mathbb{Z}^3 \rtimes_B \mathbb{Z}$ , with one real and two complex conjugate eigenvalues, none roots of unity, thus also not automatic.

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